



Long-term recovery of Mediterranean ant and bee communities after fire in southern Spain



J. Manuel Vidal-Cordero^a, Elena Angulo^{a,c}, Francisco P. Molina^a, Raphaël Boulay^b, Xim Cerdá^{a,*}

^a Estación Biológica de Doñana, Consejo Superior de Investigaciones Científicas (CSIC), Av. Amerigo Vesputio 26, 41092 Sevilla, Spain

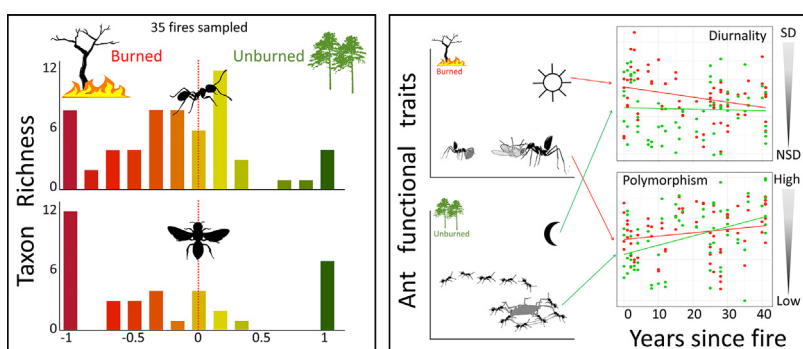
^b Institut de Recherches sur la Biologie de l'Insecte, CNRS UMR 7261, Université François Rabelais de Tours, 37200 Tours, France

^c Université Paris-Saclay, CNRS, AgroParisTech, Ecologie Systématique Evolution, Orsay, France

HIGHLIGHTS

- We focus on the effects of Mediterranean wildfires on ants and bees in South Spain.
- Fire increases overall ant and bee richness, regardless of time since fire.
- Recently burned areas harbor ants with functional traits suited to open areas.
- Over time, fire influenced some ant traits less; bee traits showed no change.
- Fire affected the taxonomic and functional composition of ant and bee communities.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Manuel Esteban Lucas-Borja

Keywords:

Wildfire
Pine forest
Formicidae
Apoidea
Functional diversity
Taxonomic diversity

ABSTRACT

Wildfires play a determinant role in the composition and structure of animal communities, especially for groups closely associated with the vegetation and soil, such as bees or ants. The effects of fire on animal communities depend on the functional traits of each group. Here, we assessed the impacts of fire and time since fire on the taxonomic and functional responses of ant and bee communities. We sampled 35 pine forests in Andalusia (southern Spain) that had experienced fire in the past (0 to 41 years ago). Specifically, we explored whether a) fire increased taxonomic and functional diversity and changed community composition in communities in the short term and b) fire influence (increase or decrease) on ant communities would be dependent on time since fire. We found that ant and bee taxonomic richness increased regardless of time since fire. Different approaches gave the same result, such as taxonomic diversity indexes (ant abundance, ant richness and ant Shannon diversity index), the changes in species richness in ant and bee communities, as well as the higher number of ant and bee species prone to the burned habitat than to the unburned habitat, using the I_{habitat} Index. Besides environmental variables (such as the effects of different *Pinus* species or elevation), time since fire changed the taxonomic composition of ant communities and the functional composition of bee communities. Moreover, six of the 13 ant functional traits explored differed between burned and unburned areas, with the degree of difference declining as time since fire increased. For example, burned areas contained ant communities with more ground-nesting species and strictly diurnal species, functional traits that are characteristic of open areas. In contrast, other traits persisted in burned areas over the long term, notably a higher degree of worker polymorphism and species monogyny. Our study shows how much short- and long-term effects of fire on ant and bee communities differ; while richness increases in the long-term, some functional traits are also filtered in the short-term. We suggest that fire in Mediterranean coniferous ecosystems could have a positive effect on these groups and should not be overlooked.

* Corresponding author.

E-mail addresses: angulo@ebd.csic.es (E. Angulo), fpaula.molina@ebd.csic.es (F.P. Molina), xim@ebd.csic.es (X. Cerdá).

1. Introduction

Fire is a crucial ecological process that simultaneously acts as a destructive disturbance force and provides a variety of ecosystem services (McLauchlan et al., 2020; Pausas and Keeley, 2019). Wildfires are the most frequent target of research on disturbance events (Viljur et al., 2022). They greatly alter forest ecosystems by reshaping the landscape and modifying the dynamics and structure of communities (Pausas and Keeley, 2009). They may also pose a global challenge to conservation efforts, making it essential to understand how biodiversity is affected by fire regimes (Kelly et al., 2020). While studies have traditionally focused on plants (Keeley et al., 2012, 2011), more recent research has explored fire's impact on animals, including arthropods (New, 2014; Yekwayo et al., 2018; Carbone et al., 2019).

Arthropods play critical roles in forest ecosystems and must be considered in any forestry management plan that seeks to maintain biodiversity (Kremen et al., 1993). This group's responses to fire have mostly been studied from a taxonomic perspective. Essentially, species abundance and richness have been found to increase rapidly after fire; they then either remain fairly constant until the next fire occurs, or their levels may begin to decline 7 to 15 years after fire occurrence (New, 2014). However, to properly understand the mechanisms behind these taxonomic responses, it is important to also examine shifts in the functional traits represented within arthropod communities (Braga et al., 2013; Mouillot et al., 2013). This approach can clarify how organisms respond to wildfire and reveal the specific adaptations that promote survival or recovery (Pausas and Parr, 2018). To this end, researchers can utilize both, data on the taxonomy of species and functional traits found within communities (Májeková et al., 2016).

Ants and bees are known to provide important ecosystem services (Del Toro et al., 2012; Kremen et al., 1993). In Mediterranean ecosystems, where wildfires are frequent (Keeley et al., 2012) both groups play a key role in ecosystem regeneration. Bees provide pollination services and thus influence the diversity of flowering plants (Galbraith et al., 2019; Koh et al., 2016). Ants have mutualistic relationships with plants, which include dispersing seeds and fruits (Barroso et al., 2013; Boulay et al., 2007; Ordóñez and Retana, 2004; Retana et al., 2004) and pollination (De Vega et al., 2014; Gómez et al., 1996). They also have antagonistic relationships; for example, they are seed predators (Arnan et al., 2010; Azcárate et al., 2005).

In general, fire changes the composition and state of Mediterranean ant and bee communities by inducing both direct and indirect mortality attributable to habitat modifications and by setting the stage for colonization and recolonization from neighboring unburned areas (Ne'eman et al., 2000; Pausas, 2019; Pausas and Parr, 2018; Wong et al., 2019). First, many ant and bee species nest in the ground, which increases their likelihood of surviving fire's extreme above-ground temperatures (Cane and Neff, 2011). However, those that nest at shallow depths or on vegetation face the direct effects of fire (fire can burn ant nests and thus kill the ants inside). Besides, survival ants face the indirect effects of fire (fire can diminish the overall resource availability by changing vegetation structure, e.g., Rosa et al., 2021). Second, community resilience is shaped by the identity and diversity of plants present before and after the fire, which provide direct and indirect resources to ants and bees (Arnan et al., 2019, 2009, 2007, 2006; Caut et al., 2014; Fagundes et al., 2015; Mola et al., 2020; Potts et al., 2003). Third, colonization dynamics are conditioned by species dispersal capacity (Pausas and Parr, 2018). In ants, dispersal distance is linked to colony foundation mode, which may be independent (i.e., a winged queen able to fly long distances) or dependent (i.e., a group of ants that travel short distances on foot) (Amor et al., 2011; Cronin et al., 2013). In contrast, in bees, dispersal distance is determined exclusively by flight capacity, which is greater for larger species (Guédot et al., 2009; Ne'eman et al., 2000). Therefore, burned habitats host a combination of species: those that survived the fire and can persist under the new environmental conditions and those that arrive from unburned areas, attracted by the available resources.

Most research looking at the taxonomic responses of bee communities to fire has found an increase in species abundance or diversity: either

because there is an increase in floral resources and boring insect exit holes used by cavity-nesting bees (Galbraith et al., 2019); because bee abundance is negatively related to canopy cover (Grundel et al., 2010); or because floral abundance persisted for longer time in burned sites (Mola and Williams, 2018; Moretti et al., 2009; Potts et al., 2003); in all these cases, the pattern that usually emerges is a rapid recovery with a peak in diversity of both flowers and bees in the first years after a fire. For ants, however, taxonomic responses have been quite variable across the world (New, 2014): in the Brazilian rupestrian complex, the ant fauna of burned areas maintained a constant species diversity along time but greatly changed species assemblage due to the occurrence of several species not found in the unburned area (Anjos et al., 2015); in the Australian pindan savanna woodland, the ant fauna was highly resilient to fire and most of the 84 ant species (except 2) did not show any positive change in abundance over time since fire (Andersen et al., 2014); but after experimental fires in a similar savanna woodland Parr and Andersen (2008) observed a dramatic decline in dominance of the patchily distributed forest-associated species *Oecophylla smaragdina* and *Papyrius* sp., although fire had no effect on overall dominance by open savanna species of *Iridomyrmex*. Yet, relatively little is known about how ants and bees respond functionally to fire over the intermediate and long term. Functional redundancy has been observed in bees: after a fire, bee abundance climbs, and while taxonomic diversity may decrease, functional diversity is maintained (Moretti et al., 2009). In Mediterranean ant communities, fire results in greater functional diversity, mainly due to the presence of rare species that are functionally unique (Arnan et al., 2019). In both cases, such changes could have major consequences for ecosystem processes and services.

Thus, to properly understand how fire impacts bee and ant communities over time, it is important to approach the topic from both a taxonomic perspective (i.e., traditional measures of biodiversity such as species abundance, species richness, and diversity measured with different indexes) and a functional perspective (i.e., indices such as functional richness and functional divergence), utilizing data on the composition and functional traits of communities (Májeková et al., 2016). A large number of studies have recently examined both functional and taxonomic diversity in animal assemblages (e.g., snakes: Adams et al., 2022; bats: López-Baucells et al., 2022; fishes: Villéger et al., 2010; birds: Morelli et al., 2018; mammals, beetles, and snakes: Guiden et al., 2021; spiders and carabids: Ridet et al., 2021; forest arthropods: Wermelinger et al., 2017; ants: Arnan et al., 2017, 2018; and bees: Villalta et al., 2022). However, research focusing specifically on fire is scarce (e.g., beetles: Moretti et al., 2010, Barber et al., 2017; ants: Arnan et al., 2013, 2020; and bees: Lazarina et al., 2016, Moretti et al., 2009).

The objective of our study was to assess the effects of fire and time since fire on two taxonomic groups—ants and bees. We conducted our research along a regional gradient in southern Spain in 35 pine forests that had experienced a wildfire at some point in the past (between 0 and 41 years ago). We characterized the forests' geographical and climatic conditions, and we assessed vegetation cover and the diversity and composition of ant and bee communities. Within each study area, we compared a burned area with a nearby unburned area. We tested the following hypotheses: (1) since fire reduces canopy and vegetation cover, and thus reduces vegetation structural complexity and the amount of feeding resources, we expected that, in the short term, fire could decrease taxonomic and functional diversity at burned versus unburned areas; alternatively, the appearance of new ecological niches (open areas in the forest) could boost taxonomic and functional diversity at burned versus unburned areas; (2) since recently burned areas tend to harbor species typical of open habitats but less recently burned areas harbor species with greater ties to vegetation, we expected that taxonomic and functional diversity would decline over time at the burned areas until they reached the values found at the unburned areas; (3) the null hypothesis for hypotheses 1 and 2 was that taxonomic and functional diversity would be related to environmental variables (i.e. latitude, longitude, elevation, climate, lithology or the pine species present previous to fire) rather than time since fire; and (4) in accordance with the above-noted changes in vegetation, we expected the taxonomic

and functional composition of ant and bee communities to be largely characterized by species typical of open habitats at recently burned areas and by species more associated with vegetation at less recently burned areas.

2. Methods

2.1. Study areas and fire effect variables

To select our study areas, we used databases provided by the Environmental Information Network of Andalusia (Junta de Andalucía REDIAM, 2022). Specifically, we employed two databases that contain records of fires in the region: one contains information for the period between 1975 and 2008 and the other for the period between 2008 and 2013. Both were merged with a database with information on land use since 1956. Between 1975 and 2013, a total of 56,289 fires were recorded in Andalusia. Using geographic information systems (QGIS), we preliminarily selected areas in which older conifer forests had burned ($n = 3614$). To obtain as much homogeneity as possible among study areas, we further filtered the results based on the following criteria: 1) the burned area had to exceed 4000 ha; 2) the wood had not to be removed; 3) the distance from the center of the burned area to the edge had to exceed 200 m; 4) suitable unburned (control) areas must exist nearby; 5) the burned area had to have experienced a single fire. We thus obtained a list of 95 burned areas that we visited during the spring and summer of 2015 to assess ease of sampling (e.g., access) and comparability of the burned areas with their unburned counterparts (e.g., similar slope, orientation). Access was the criteria that finally discarded most of the 95 areas, as many times burned areas had no available roads or paths to get there or there were unavailable sampling places due to steep slopes or fallen trees. Thus, we finally obtained a total of 35 study areas that had experienced fire at different years in the past (Fig. 1; Table S.1).

To study the effect of fire, we used different variables. The “fire effect” distinguished between burned and unburned areas. We used a paired design with a space-for-time substitution approach, in which each burned area (the treatment) had a paired unburned area (the control), which

represents the native community of the burned area before the fire. The control areas were between 275 and 2900 m from their paired burned areas (mean of 691 m). To take into account when the area experienced fire, we used the “time since fire” variable that represents how long ago the fire occurred (0 to 41 years) and could be used as a continuous variable or a categorical one (see next section).

2.2. Vegetation cover

The 35 study areas were all found in pine forests (*Pinus halepensis*, *P. pinaster*, *P. nigra*, and *P. pinea*; see Table S1) that had a relatively dense understory composed of typical Mediterranean shrubs, including *Salvia rosmarinus*, *Pistacea lentiscus*, *Juniperus oxycedrus*, *Chamaerops humilis*, *Halimium* spp., *Lavandula* spp., *Cistus* spp., *Ulex* spp., *Erica* spp., and a broad variety of herbaceous plants (mainly members of Apiaceae, Asteraceae, and Poaceae). To characterize the structure of the vegetation in the study areas, we established two transects at each of the burned and unburned areas; within an area, the two transects were separated by 200 m minimum. The transects were 75 m long and 10 m wide. Every five meters, we recorded the presence or absence of four strata: herbaceous plants, small shrubs, large shrubs, and trees, noting species identity (Fig. 1).

We calculated the value of a plant cover index (Veg) for each transect using the presence/absence data (Dataset S1, spreadsheet *data.veg*). This value ranged between 0 and 100 %, where 100 % meant the four strata were present at all fifteen sampling points, while 0 meant the four strata were absent at all fifteen sampling points.

We performed a preliminary analysis to clarify two issues: did vegetation cover vary across the burned areas in association with time since fire? Was vegetation cover similar across all the unburned areas? To answer these questions, we carried out linear models using the *lme* function (*nlme4* package; Bates et al., 2015). One model was performed in which the dependent variable was the Veg index and the independent variables were fire (burned vs. unburned), time since fire (continuous: 0 to 41 years), and their interaction. Four additional models had the same independent variables, but the dependent variable was the mean presence of each stratum

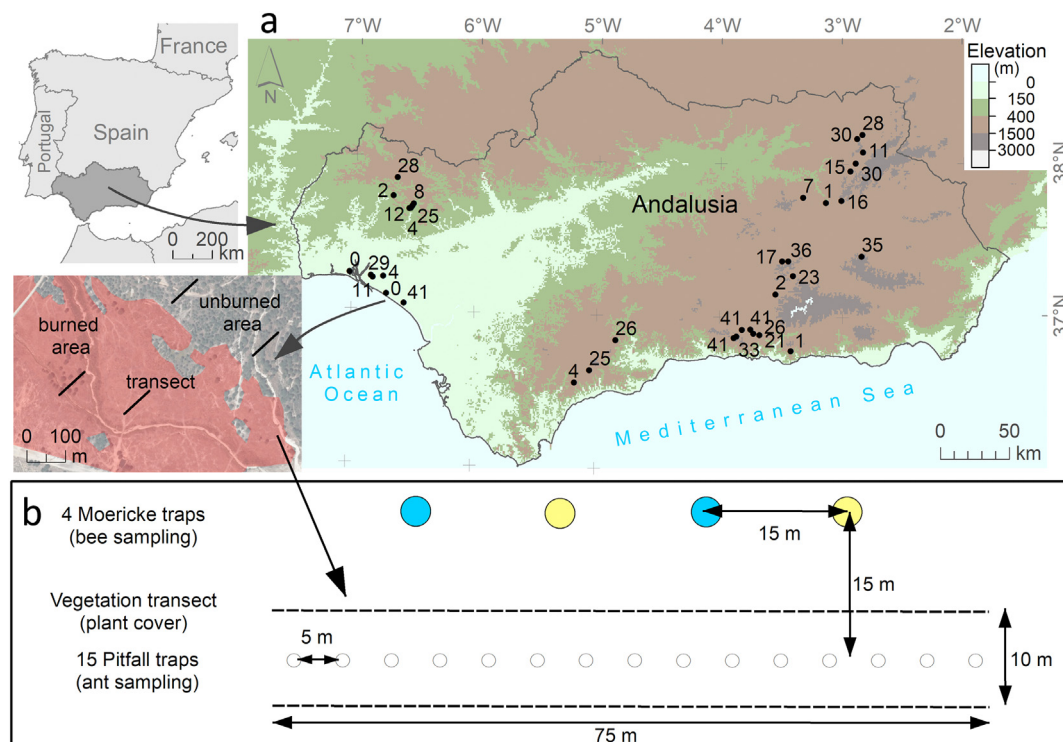


Fig. 1. (a) Map of the study areas and (b) diagram of the sampling protocol. The black dots on the map are the sampling areas, and the numbers indicate the time since the fire (years).

across the 15 points sampled. As we expected, at burned areas, vegetation cover differed significantly with time since fire, except for the herbaceous strata (Table S2a). In contrast, there were no differences associated with time since fire among any of the strata at unburned areas (Table S2b). The interaction between fire and time since fire was significant for all the strata with the exception of the herbaceous plants, which indicates that this stratum does not provide information about the area's recovery from fire (Table S2c, Fig. S1).

Given the importance of plant community recovery in ant and bee community recovery, we explored whether vegetation cover formed natural groups based on time since fire. To this end, we performed a cluster analysis using the *hclust* function (*cluster* package; Maechler et al., 2022). The dependent variable was the vegetation cover at the burned areas for the three strata found to be significant above. The results revealed the presence of three categories (Fig. S2): category 1—fire occurred 0 to 4 years ago; category 2—fire occurred 7 to 12 years ago; and category 3—fire occurred 15 to 41 years ago. These categories are similar to those used by Peralta et al. (2017) when studying fire's effects on plant-bee networks.

2.3. Ant and bee sampling

Sampling of ants and bees was conducted in 2016, from May through August. The goal was not to comprehensively sample ant and bee diversity, but rather to rapidly assess group presence at burned and unburned areas for the sake of comparison. We therefore used the most common and straightforward sampling method for each group: pitfall traps for ants (Retana and Cerdá, 2000; Ward et al., 2001) and Moericke pan traps for bees (Bogusch et al., 2015; Grundel et al., 2010). For the ants, the open pitfall traps were 220-cm³ plastic cups that were 2/3 filled with soapy water. For the bees, the Moericke pan traps were 1000-cm³ yellow and blue plastic dishes that were filled with soapy water. Bee species trapped in Moericke pan traps are normally biased to larger bees or those that fly higher in the vegetation (Prendergast et al., 2020). In each study area, we placed 60 pitfall traps (i.e., 60 traps × 35 study areas) and 16 Moericke traps (i.e., 16 traps × 30 study areas) along 4 transects (2 at the burned area and 2 at the unburned area). Each transect contained 15 pitfall traps separated by 5 m; along a parallel line were 4 Moericke traps (2 yellow and 2 blue in an alternating sequence) separated by 15 m (Fig. 1).

The pitfall traps and Moericke traps were run for 48 h. Any biological material captured was stored in 70 % alcohol until identification could occur. Ants and bees were sorted out from the rest of the invertebrates. Ants were identified to species level; bees were identified to genus level, or to species level whenever possible (Dataset S2). Taxonomic identification was carried out by the two taxonomist authors (JM Vidal-Cordero and FP Molina) using their own knowledge and the guides (see Appendix S1). Taxonomic doubts were sent to experts (ants: Joaquín L. Reyes-López; bees: Luis Óscar Aguado Martín) to confirm the identification. In the following analyses, we used species-level information for ants. For bees, we used either genus- or species-level information, depending on how much taxonomic resolution was possible.

2.4. Life-history traits

We characterized 13 and 6 functional traits for ants and bees, respectively. These traits are associated with three important processes (survival, persistence, and colonization) that play a crucial role in autecology and ecosystem functioning and that offer up a potential mechanistic explanation for how fire affects both ants (Arnan et al., 2013) and bees (Grundel et al., 2010; Love and Cane, 2016).

First, we looked traits related to surviving fire, notably nesting site (ants and bees nesting in the ground rather than in the vegetation, wood, snail shells, or hives) (Tables S3, S4). Next, we examined traits related to persistence after fire. For ants, according with Arnan et al. (2020) and Vidal-Cordero et al. (2022), the traits allowing to persist after fire are: behavioral dominance (dominants, following Arnan et al., 2012; Cerdá et al., 2013 have a higher probability of persistence), body size (larger ants), colony

size (larger colony size), nest number (polydomous, with more than one nest per colony), queen number (polygynous, more than one queen per colony), diurnality (strictly diurnal), foraging strategy (collective), diet type (insects or seeds, highly available after fire), and degree of worker polymorphism (more pronounced worker polymorphism) (Table S3). For bees, these traits were body size (four categories) and intertegular distance (wildfires may select for larger bees, Burkle et al., 2019), degree of floral specialization (polylectic or oligolectic; e.g. oligoleges are more abundant in open areas, Grundel et al., 2010), and tongue length (short-tongued bees are more frequent in recently burned areas, Moretti et al., 2009) (Table S4). Finally, we explored traits related to the colonization of burned areas, which were colony foundation type (dependent, independent, or both) for ants and lifestyle (solitary, social, or parasitic) for bees (Tables S3, S4).

Information about functional traits was mostly obtained from the literature (Appendix S1). Data on ant traits came from Arnan et al. (2013) and Retana et al. (2015) (Table S3, Dataset S1, spreadsheet *ant.traits*). Data on bee traits were gathered via an extensive search of public databases and the scientific literature (Table S4, Dataset S1, spreadsheet *bee.traits*). We also measured the intertegular distance of the bees we captured using a micrometer and stereomicroscope. For some species of both groups, we additionally drew on unpublished data provided by experts (ants: Joaquín L. Reyes-López; bees: Luis Óscar Aguado Martín).

2.5. Taxonomic diversity

All the analyses were performed using R (v. 4.1.3; R Core Team, 2022). First, we characterized the taxonomic structure and composition of the ant and bee communities. We defined ant abundance (A_{ants}) as the mean number of ants per transect ($N = 15$) using the total number of ants in each trap in the calculation. We defined bee abundance (A_{bees}) as the total number of bees captured per transect. When traps were missing, we extrapolated ant and bee abundance data to the total number of traps for each transect (for ants, 103 traps across 46 transects went missing; for bees, 11 traps across 8 transects went missing; total transects = 140, Dataset S2). For each transect, we calculated the following taxonomic diversity indices (Dataset S1, spreadsheets *data.ants* and *data.bees*): (1) observed taxon richness (S_{ants} and S_{bees}), which was the total number of species/genera observed per transect; (2) the Shannon diversity index (H_{ants} and H_{bees}), which accounts for both the number of species/genera and their relative abundance; and (3) Pielou's evenness (J_{ants} and J_{bees}), which is the Shannon diversity index divided by the natural logarithm of the observed taxon richness. S and H were calculated using the *estimate* function (*vegan* package, Oksanen et al., 2018).

2.6. Functional diversity

To characterize the functional composition of ant and bee communities, we used two widely recognized approaches in ecological research: 1) the community weighted mean (CWM) of traits, which indicates a community's most common traits after accounting for differences in taxon abundance (Laliberté and Legendre, 2010) and 2) the functional diversity (FD) in traits, which indicates how much species within a community vary in their traits (Arnan et al., 2013; Ricotta and Moretti, 2011). CWM and FD values were calculated for each transect.

The method for calculating CWM differs depending on whether traits are continuous or categorical. For continuous traits, CWM corresponds to the abundance-weighted mean and was determined as follows:

$$\bar{x} = \sum_{i=1}^s p_i x_i$$

where p_i is the relative abundance of species (or genus) i and x_i is the trait value for species (or genus) i . For categorical traits, CWM corresponds to the relative abundance of a species (or genera) associated with each trait category.

Different aspects of functional diversity have the potential to reveal the processes structuring biological communities (Mouchet et al., 2010). FD was thus broken down into four subindices that can help clarify the mechanistic effects of functional diversity on ecosystems: functional richness (FRic), functional evenness (FEve), functional dispersion (FDis), and functional divergence (FDiv). FRic expresses the amount of functional space filled by the community, but it does not account for relative abundances. Consequently, we can use FEve, which describes the evenness of abundance distribution in a functional trait space (Mason et al., 2005). However, FEve does not reveal the dispersion of species in trait space. The next step is therefore to calculate FDis (Laliberté and Legendre, 2010), which is the mean distance of individual species to the centroid of all species in multidimensional trait space. Finally, FDiv measures how abundances tend to be on the outer margins of the functional space while controlling for functional richness. These four metrics are independent of each other and of species number, evenness, and taxonomic diversity. We calculated CWM with the *functcomp* function and the four FD subindices with the *dbFD* function (FD package, Laliberté and Legendre, 2010) (Dataset S1, spreadsheets *data.ants* and *data.bees*).

2.7. Environmental variables

Given the regional scale of our study, we gathered information on the environmental characteristics of each study area. The resulting set of variables was used as covariables in the analyses (Dataset S1, spreadsheets *data.ants* and *data.bees*). This list comprised latitude, longitude, and elevation (Lat, Long, and Elev) as determined from GPS coordinates; lithological types (igneous, sedimentary, and metamorphic rock) (CSIC-IARA, 1988), as different type of soil affect ant richness and bee nesting for ground-dwelling bees (Angulo et al., 2016; Lybrand et al., 2020; Antoine and Forrest, 2021); pine species (*Pine halepensis*, *P. nigra*, *P. pinaster*, and *P. pinea*) because pines can differ in their responses to fire (He et al., 2012); and eight bioclimatic variables (WorldClim database v. 2.0— www.worldclim.org; Hijmans et al., 2005). The latter covered the period from 1970 to 2000, and the values had a spatial resolution of about 1 km². Using principal components analysis (PCA), we reduced them to two uncorrelated components that accounted for 92.3 % of the original variation (Clim. 1 and Clim. 2; Table S5).

2.8. Exploring the effects of fire

2.8.1. Rarefaction curves in burned and unburned areas

To determine whether we had sampled most of the species/genera in the study areas, we calculated abundance-based rarefaction curves for the ants and bees. We utilized the total number of observations of each taxon (species/genus) for each group (ants and bees) at the burned and unburned areas. The curves were obtained via the *specaccum* function (*vegan* package, Oksanen et al., 2018).

2.8.2. Effects of fire on taxon richness

We wanted to compare taxon richness (species/genus) at burned vs unburned areas. First, we calculated the difference in ant and bee occurrence between burned and unburned areas using a habitat use index, modified from Suggett et al. (2012): $I_{\text{habitat}} = [(2 * \text{unburned occurrence values}) / (\text{unburned occurrence values} + \text{burned occurrence values})] - 1$. The value of the index ranges from -1 to 1 , where negative values indicate more frequent association with burned habitats, and positive values denote taxa more associated with unburned habitats.

We also conducted this comparison by determining the absolute difference in taxon richness (species/genus) between the burned and unburned transects for each area study. The difference in richness was the mean richness value of the burned transects minus the mean richness value of unburned transects. We explored whether the resulting value was correlated with time since fire, accounting for environmental variables. We used one linear model (*lm* function, *stats* package; Pinheiro and Bates, 2022,

Pinheiro and Bates, 2000) for the ants and one for the bees. The difference in richness was the dependent variable, time since fire (continuous: 0 to 41) was a fixed effect, and the environmental variables were included as covariables.

We obtained a map of the differences in richness across the study areas to gain insight into the relevance of the spatial variables that could be affecting (such as latitude, longitude or altitude). We used the interpolation tool in ArcGIS 10.7, which employs the inverse distance weighted (IDW), with a neighbor search algorithm implemented with smooth circulars. This method is one of the most flexible interpolation methods. It assigns predicted values to unknown points by using surrounding observed values from a set of known points obtained from our sampling. Each measured point (the calculated difference in richness) has a local influence that diminishes with distance (ArcGis Pro 3.0). In our case, IDW interpolation assumes that the influence of the values of observed richness differences on the predicted values diminish as a function of geographic distance, with closer values having a greater influence on the predicted values than those farther away.

Finally, we examined the patterns of occurrence for each ant species across the three time-since-fire clusters for the burned and unburned areas. We calculated the average time since fire (and standard error) for the burned and unburned areas.

2.8.3. Effects of fire on taxonomic and functional community structure

We examined the effects of fire and time since fire on the taxonomic and functional community structure indexes (indices: A, S, H, J, FRic, FEve, FDis and FDiv). The ant and bee data were analyzed separately using linear mixed-effects models (*lme* function, *nlme* package; Pinheiro and Bates, 2022, Pinheiro and Bates, 2000). In the models, the dependent variable was one of the indices; fire (burned and unburned), time since fire (continuous: 0 to 41), and their interaction were fixed effects. The environmental variables were covariates, and study area was a random factor to account for the fact that different transects were sampled within the same study area. To meet statistical assumptions regarding normality, homogeneity and heteroscedasticity, we transformed the dependent variables using the transformation that better suit normality assumptions in each case (see Results, Table 1).

2.8.4. Effects of fire on taxonomic and functional community composition

We examined the effects of fire and time since fire on the taxonomic and functional community composition of ants and bees. To this end, we created matrices that combined information on taxonomic and functional composition for each transect. For the taxonomic analysis, the matrix cell values were ant occurrence (the number of traps in which each species occurred per transect) and bee abundance (the number of individuals of each species/genus per transect). For the functional analysis, we used the abundance-weighted trait values. We ran PERMANOVA tests using the *adonis* function (*vegan* package, Oksanen et al., 2018). Fire (burned and unburned), time since fire (continuous: 0 to 41 years), and their interaction were fixed effects. Study area was added as a random factor to account for the fact that different transects were sampled within the same study area. We used non-metric multidimensional scaling (NMDS) based on Bray-Curtis dissimilarities to visually explore how the taxonomic and functional composition metrics were associated with fire (burned and unburned) or time since fire (three clusters).

2.8.5. Effects of fire on functional traits

While the community-level analysis allowed us to evaluate different facets of the functional space occupied by entire communities, comprising multiple traits, we were also interested in determining whether individual traits were associated with fire or time since fire. Thus, we tested the influence of fire (burned and unburned), time since fire (continuous: 0 to 41), and their interaction on each of the functional traits of ants and bees. We used linear mixed-effects models that included study area as a random factor. We transformed the dependent variables to achieve normality whenever required.

Table 1

Effects of fire and time since fire on the taxonomic community structure of ants (a) and bees (b); and on the functional community structure of ants (c) and bees (d). Metrics are A = abundance, S = richness, H = Shannon diversity index, and J = Pielou's evenness; FRic: functional richness; FEve: functional evenness; FDis: functional dispersion; and FDiv: functional divergence. The abbreviations Log, ², and ^{1/2} indicate the use of a logarithmic, squared, or square-root transformation, respectively. Fire: fire treatment (burned and unburned); Time: how long ago the fire occurred (0 to 41 years); FxT: the interaction between the former two variables. Statistically significant results are in bold. The complete models are in Table S9, S10.

a) Ants		Log(A)		S		H ²		J ²	
	DF	F	P	F	P	F	P	F	P
Fire	1, 101	9.36	0.003	9.54	0.003	6.62	0.011	0.37	0.547
Time	1, 23	0.06	0.815	0.44	0.515	0.64	0.431	0.16	0.693
F x T	1, 101	1.81	0.182	0.08	0.772	0.35	0.557	2.01	0.159
b) Bees		Log(A)		S		H ²		J ²	
	DF	F	P	F	P	F	P	F	P
Fire	1, 51	2.83	0.099	2.19	0.145	0.58	0.451	2.45	0.130
Time	1, 17	6.16	0.024	0.21	0.651	0.43	0.522	2.53	0.143
F x T	1, 51	0.20	0.653	0.10	0.753	0.80	0.375	2.61	0.119
c) Ants		FRic		FEve		FDis ²		FDiv	
	DF	F	P	F	P	F	P	F	P
Fire	1, 101	2.97	0.088	0.36	0.551	0.07	0.788	9.63	0.002
Time	1, 23	2.44	0.132	0.00	0.952	4.01	0.057	3.26	0.084
F x T	1, 101	0.05	0.832	0.04	0.838	0.31	0.581	0.01	0.934
d) Bees		Log(FRic)		FEve		FDis ^{1/2}		FDiv	
	DF	F	P	F	P	F	P	F	P
Fire	1, 51	0.03	0.882	3.86	0.188	1.27	0.265	0.41	0.586
Time	1, 17	0.00	0.992	0.07	0.813	2.33	0.145	0.06	0.828
F x T	1, 51	1.09	0.406	3.21	0.215	0.69	0.411	5.57	0.142

3. Results

3.1. Effects of fire on community richness

We captured a total of 71,050 ant workers belonging to 61 species, 23 genera, and 5 subfamilies (Table S6). There were 322 individual bees from 37 species, 24 genera, and 6 families (Table S7).

For the ants, the species accumulation curves were nearly asymptotic in both burned and unburned areas (Fig. 2a), suggesting that we had sampled most of the species present in the study areas. However, for the bees, the curves did not approach an asymptote, indicating that we had not comprehensively characterized this group (Fig. 2b). Interestingly, for both ants and bees, the curve for the unburned area lay below the curve for the burned area, indicating that richness was greater in burned areas.

We found further support for this pattern when we analyzed ant and bee occurrence: out of the 61 ant species, 56 % had negative I_{habitat} values, and 34 % had positive values, indicating that ants were more likely to be found at burned areas (Fig. 2c). Similarly, out of the 37 bee species, 62 % had negative I_{habitat} values, and 27 % had positive values, indicating that bees were more likely to occur at burned areas (Fig. 2d). It is important to note that burned areas were more open habitat.

Additionally, when differences in species richness were mapped across the study areas (Fig. 3), the same result was seen. The richness of ant and bee communities was greater at burned than unburned areas (more red vs. blue), a pattern that was seemingly more pronounced in ants. However time since fire did not significantly affect the differences in species richness ($F_{1,22} = 0.00$, $p = 0.948$ for ants; $F_{1,22} = 1.56$, $p = 0.228$ for bees). Out of all the environmental variables, only latitude was correlated with differences in community richness and only in the case of ants ($F_{1,22} = 5.26$, $p = 0.03$) (Table S8).

Finally, when we examined the relationship between ant species occurrence and time since fire, we observed differential responses among species

(Fig. 4). The most frequently occurring ant species (on the top of Fig. 4) were found at burned areas that had experienced fire >15 years ago. Some frequently occurring species (occurrence >10, on the center of Fig. 4) were found at areas that had burned less recently (15 to 41 years ago) as well as moderately recently (4 to 12 years ago): *Aphaenogaster senilis*, *Messor bouvieri*, *Cataglyphis tartessica*, *Messor barbarus*, *Cataglyphis rosenhaueri*, *Goniomma kugleri*, *Oxyopomyrmex saulcyi*, and *Messor maroccanus*. Three species, *Cataglyphis hispanica*, *Tetramorium forte*, and *Messor lusitanicus* tended to occur at areas of moderately recent burns (4 to 12 years ago), while *Temnothorax gredosi* was most common at recently burned areas (0 to 4 years ago). The other ant species associated with recent burns occurred at extremely low levels (occurrences <3, on the bottom of Fig. 4).

3.2. Effects of fire on taxonomic community structure

Ant abundance, ant richness, and ant diversity were significantly greater at burned versus unburned areas (A_{ants} , S_{ants} , H_{ants} ; Table 1a, Fig. 5a,b,c). For bees, there was the same trend in all three metrics (A_{bees} , S_{bees} , H_{bees} and J_{bees}), but it was not statistically significant (Table 1b), except bee abundance, which was significantly higher at areas that had burned more recently (Fig. S3a).

That said, the interaction between fire and time since fire was not significant for any of the taxonomic indices for either ants or bees (Table 1a,b). Some of the taxonomic indices were associated with certain environmental variables (Table S9). S_{ants} , H_{ants} , and J_{ants} were significantly lower at higher latitudes but significantly higher at higher elevations. A_{ants} was negatively correlated with vegetation cover. Finally, A_{ants} and J_{ants} were impacted differently by the presence of pines: A_{ants} was highest in association with *P. pinaster* and *P. halepensis*, while J_{ants} was highest in association with *P. nigra* (Table S9, Fig. S3b,c).

3.3. Effects of fire on functional community structure

In ants, there were a few key results with regards to functional community structure. $FDiv_{\text{ants}}$ was higher at unburned versus burned areas (Table 1c, Fig. 5d), and $FDis_{\text{ants}}$ was lower for more recent fires (Table 1c, Fig. S4a). That said, the interaction between fire and time since fire was not significant for any of the functional indices examined in ants. Some of the functional indices were influenced by environmental variables (Table S10): *P. pinaster* and *P. halepensis* were associated with the highest $FDis_{\text{ants}}$ values (Fig. S4b), while $FRic_{\text{ants}}$ and $FDis_{\text{ants}}$ increased with elevation (Table S10, Fig. S4c,d). In bees, there were no significant results with regards to functional community structure and fire (Table 1d). We only observed a negative relationship between $FDis_{\text{bees}}$ and the Clim. 1 component (Table S10, Fig. S4e).

3.4. Effects of fire on taxonomic and functional community composition

The taxonomic community composition of ants was significantly influenced by fire, time since fire, and their interaction (fire: $F = 2.38$, $p = 0.001$, $R^2 = 0.02$; time: $F = 3.98$, $p = 0.001$, $R^2 = 0.05$; and interaction: $F = 1.09$, $p = 0.002$, $R^2 = 0.01$; Fig. 6a–c). Ant communities at burned areas occupied a smaller and more distinct area in NMDS space than did ant communities at unburned areas, especially when the fire was moderately to less recent (7–12 years ago and 15–41 years ago; Fig. 6b,c). This pattern was not seen for the most recent fires (0–4 years ago), which could explain the interaction (Fig. 6a). This can be seen by a reduction of the overlapping of the two ellipses (burned and unburned ant communities) from Fig. 6a–c. Within the burned areas, there was more community similarity in the case of moderately to less recent fires than for recent fires and less recent fires (Fig. S5a,b).

However, for bees, only the fire treatment significantly affected community composition ($F = 1.55$, $p = 0.02$, $R^2 = 0.02$, Fig. S5c); there was no effect of time since fire ($F = 3.60$, $p = 0.07$) or the interaction ($F = 0.72$, $p = 0.34$). Bee communities in burned areas although highly

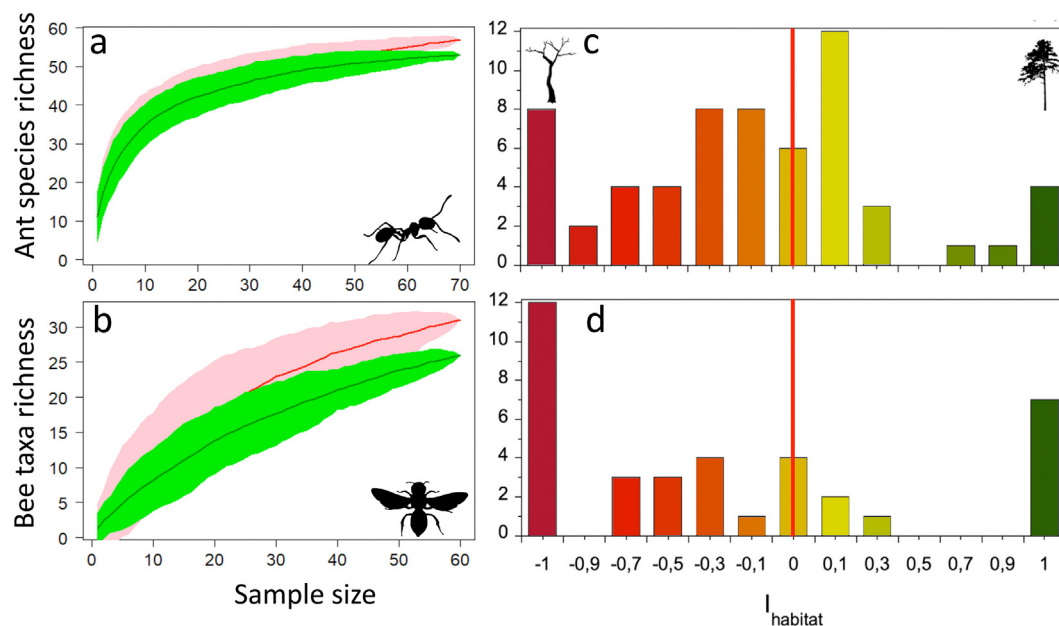


Fig. 2. Rarefaction curves for (a) ants and (b) bees and distribution of I_{habitat} values for (c) ants and (d) bees at burned and unburned areas. Ants were sampled using pitfall traps, and bees were sampled using Moericke traps. Occurrence data were used for ants, while abundance data were used for bees. The thick lines in (a) and (b) show the observed taxa richness for ants and bees; the red curves indicate the burned areas, and the green curves indicate the unburned areas; the shaded area is the standard error.

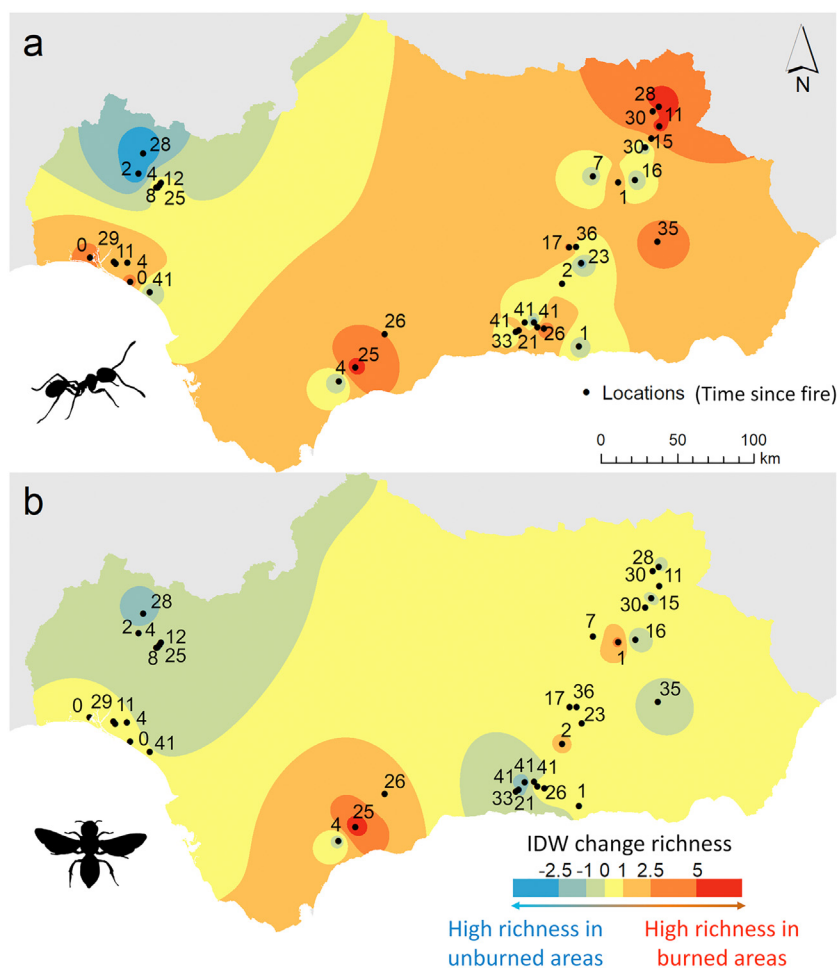


Fig. 3. Differences in richness of (a) ant species and (b) bee taxa between burned and unburned areas. For each study group and area, the difference in richness was the difference between the mean richness value of the burned transects and the mean richness value of unburned transects (i.e., positive number = greater richness in the burned transects). The black dots on the map are the study areas, and the numbers indicate the time since the fire (years).

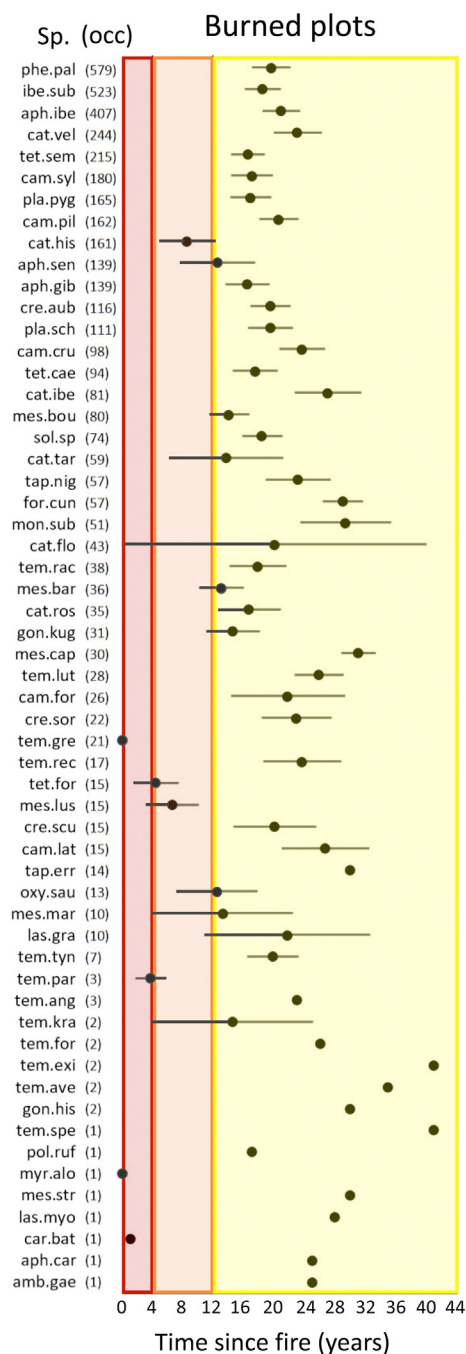


Fig. 4. Ant species occurrence and time since fire in burned plots. The species are sorted by decreasing order of occurrence (mean $\pm$ SE). On the left axis is the total number of occurrences for each species. In red are the most recently burned areas (category 1: fire occurred 0 to 4 years ago). In orange are the moderately recently burned areas (category 2: fire occurred 7 to 12 years ago). In yellow are the least recently burned areas (category 3: fire occurred 15 to 41 years ago). Occurrence is the number of pitfall traps containing a given species. The abbreviations are listed in Table S6.

overlapping in NMDS space with the communities in unburned areas, occupied a smaller area (as in ant communities).

There were no significant effects of fire on the functional composition of ant communities (fire: $F = 0.24$, $p = 0.58$; time: $F = 2.88$, $p = 0.16$; and interaction: $F = 1.53$, $p = 0.09$). The same was mostly true for bees (fire: $F = 0.94$, $p = 0.42$ and time: $F = 3.66$, $p = 0.20$; and interaction: $F = 1.18$, $p = 0.08$, $R^2 = 0.03$) (Fig. 6d–f). We observed that the functional communities of bees at burned areas occupied a larger and more distinct

area in NMDS space than did functional communities of bees at unburned areas (Fig. 6d–f).

3.5. Effects of fire on functional traits

For ants, six of the 13 functional traits analyzed were affected by the interaction between fire and time since fire (Table S11): differences between burned and unburned areas were more pronounced for more recent fires (Fig. 7). At recently burned areas, there were more ground-nesting species (Fig. 7a), more strictly diurnal species (Fig. 7b), species with greater worker polymorphism (Fig. 7c), more monogynous species (Fig. 7d), more species with dependent colony foundation (Fig. 7e), and species that consumed seeds (Fig. 7f). These differences became less dramatic as the time since the fire increased. Other ant traits were affected only by fire (independently of time since fire): burned areas hosted more monodominant species (Fig. S6a), more species that use individual foraging (Fig. S6b), larger species (Fig. S6c), and species that consumed no insects (Fig. S6d).

In contrast, for bees, only two of the six traits (nesting site and floral specialization) were affected by the interaction between fire and time since fire (Table S11), and they displayed more opaque patterns.

4. Discussion

In this study, we explored the taxonomic and functional responses of ant and bee communities to wildfires in Andalusian pine forests that had burned at various moments in the past. Our main finding was that values of most of the taxonomic indices (species richness, abundance, and taxonomic diversity) and of functional divergence were higher at burned areas, at least for ants. Fire also shaped the taxonomic community composition. An effect of fire treatment was seen in both ants and bees. There was also an impact of time since fire in ants. Although neither the functional indices (except for functional divergence) nor functional composition responded in a clear way to fire, we did observe pronounced differences in six ants' functional traits. These differences were largest at the most recently burned areas and then became less dramatic with greater time since fire. Our results confirm that fire alters ant communities more than bee communities; firstly taxonomically, increasing ant richness, abundance and diversity at the long term; and secondly functionally, with some ant traits altered in the short term or persisting over the long term. All these short- and long-term changes seem to suggest that fire in Mediterranean coniferous ecosystems is not as negative as one might expect and even could have a positive effect specially on some facets of ant communities.

4.1. Do taxonomic and functional diversity increase immediately after fire?

Our first hypothesis was that the taxonomic and functional diversity of ants and bees would be higher at burned areas, where fire creates new niches, or lower because fire simplifies the complexity of the ecosystem. We found that richness of both ants and bees was higher at burned areas, and ant abundance and ant diversity were significantly higher at burned areas, independently of the time since fire. The fire clears out existing vegetation and favors the appearance of ant species that generally occur in open areas (Arman et al., 2006; Rodrigo and Retana, 2006). It is thanks to these new arrivals, consisting on species living in open areas, plus the resident species that survived the fire, that ant species richness and diversity are greater at burned areas.

For bees, it has been extensively shown that communities increase in species richness and abundance in the first years after a fire (Galbraith et al., 2019; Grundel et al., 2010; Mason et al., 2021; Mola and Williams, 2018; Moretti et al., 2009; Petanidou and Ellis, 1996; Potts et al., 2003; Simmons and Bossart, 2020). Fire's removal of the vegetation promotes the appearance of a herbaceous understory that includes flowering plants, thus boosting food availability (Burkle et al., 2019). The greater resource offered by extrafloral nectaries of plants after fire could also benefit interacting ant species, as has been showed by da Silva et al. (2020). The bee community can also be recomposed in burned areas due to the

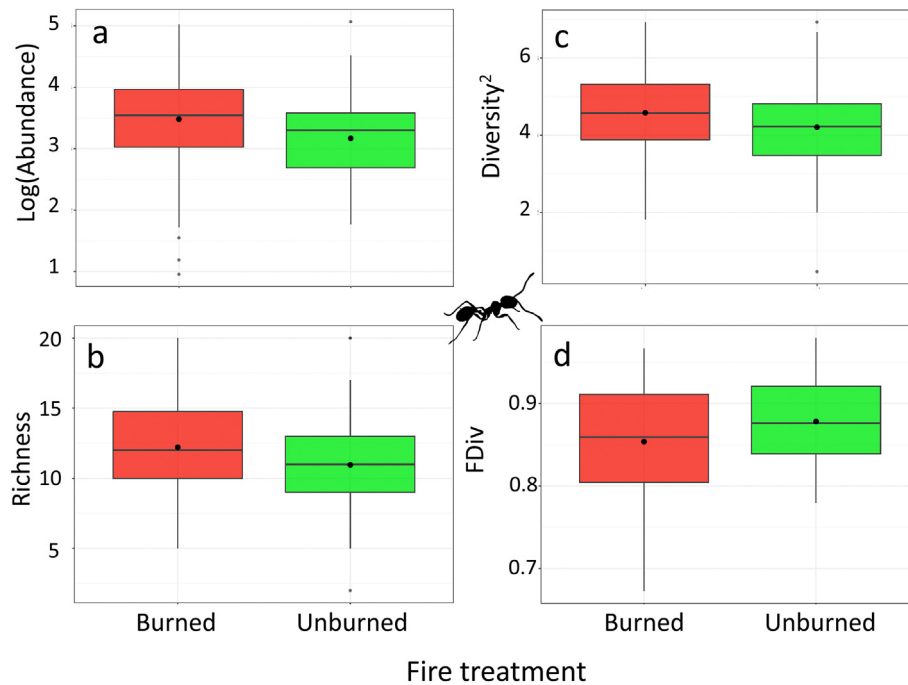


Fig. 5. Effects of fire on the taxonomic and functional community structure of ants. (a) Mean ($\pm$ SE) log-transformed ant abundance for burned versus unburned areas as well as (b) ant species richness, (c) square-transformed Shannon diversity, and (d) mean ($\pm$ SE) ant functional divergence for burned versus unburned areas. In red are the burned areas, and in green are the unburned areas. Only statistically significant results are presented (i.e. only some of the explored indices for ants, see Table 1 and Tables S8–S10).

movement of species in the metacommunity (Perillo et al., 2020). Although we did not observe significant differences between burned and unburned areas for bees, it is possible that this new resource pool could augment bee visits and abundance.

4.2. Do taxonomic and functional diversity decrease progressively with time since fire?

Our second hypothesis was that the taxonomic and functional structure of the ant and bee communities at the burned and unburned areas would become more similar the longer the time since fire. After 10 to 15 years have passed since fire, bee richness and abundance decline dramatically (Carbone et al., 2019; Potts et al., 2003). Here, we observed a significant decrease in bee abundance with time since fire, but this pattern did not extend to the other taxonomic or functional metrics of bee communities.

Ants display much more heterogeneous responses to fire than bees, as illustrated by research conducted across the globe (New, 2014). We show that fire had a constant significant effect on different taxonomic and functional diversity indexes, regardless of how much time had passed. In contrast, other systems seem to spring back quite quickly, not only because the habitat are different, but also because the ecological effects result from experimental or prescribed fires. For example, in certain US oak forests, ant communities rapidly recovered their taxonomic structure after a prescribed fire, just one month later (Verble-Pearson and Yanoviak, 2014). Similarly, in tropical Australia, savanna ant communities recovered within four months (Parr and Andersen, 2008), and African savanna communities recovered within eight months (Parr et al., 2004). Dynamics took yet another course in different Brazilian ecosystems: in the rupestrian ecosystem, the level of diversity was constant at burned sites even if the ant species assemblage changed dramatically (Anjos et al., 2015); in the Pantanal, only ten days after fire, species composition substantially changed and increased species richness, but these effects decreased at one year and disappeared after four years (Arruda et al., 2022). The Brazilian Cerrado, which has a highly diverse and specialized fauna of arboreal ants, is very susceptible to the direct effects of fire: burned trees showed a strong decline in species richness and marked changes in species composition and no sign of community recovery after two years of the last fire event

(Rosa et al., 2021). But fire was not strong enough to modify the whole structure of the ant-plant networks between ants and plants with extrafloral nectaries (Fagundes et al., 2018). However, in *campo rupestre* (Brazilian savanna), fire impacts on ant-plant interactions were evident only one month post-fire, when burned plots had lower interactions diversity and frequency than unburned plots; these differences between burned and unburned plots were no longer observed four months post-fire (Costa et al., 2018). In Mediterranean *P. nigra* forests, the taxonomic composition of ant communities had not returned to pre-fire conditions even after 19 years, which was also the case for the vegetation (Rodrigo and Retana, 2006). Finally, it is also possible that disturbance events in natural ecosystems can trigger a decrease in the functional structure of ant and bee communities (Arman et al., 2018; Bihn et al., 2010; Rader et al., 2014). Here, we found no effect of time since fire on the functional diversity of either ants or bees.

4.3. Do environmental variables or time since fire better explain taxonomic and functional diversity?

Our third hypothesis postulated that differences in taxonomic and functional diversity between burned and unburned areas could be explained by environmental variables if they were not explained by the fire. Elevation, altitude or the pine species modified some taxonomic indices (richness, diversity and evenness) and functional diversity in ants, while climate modified functional diversity in ants. This is in agreement with previous works that show that in ants taxonomic diversity increased with elevation (Bishop et al., 2014; Karaman, 2011; Sanders, 2002) although temperature can inverse this relationship (Fontanilla et al., 2019; Siqueira de Castro et al., 2020); even functional traits such as body size (Shik et al., 2019) may be related to elevation.

The plant species occurring in the area that burned provide resources and microhabitat for a variety of ants. Consequently, the forest's composition before the fire also has an influence on how well ant communities recover over the intermediate term (Arman et al., 2006). Not all pine forests have the same type of fire adaptations (He et al., 2012; Keeley, 2012). In zones with *P. pinaster* and *P. halepensis*, pine forests can eventually re-emerge after fire. These pines produce serotinous cones, which need to be exposed to fire to open and release their seeds (Elvira et al., 2021;

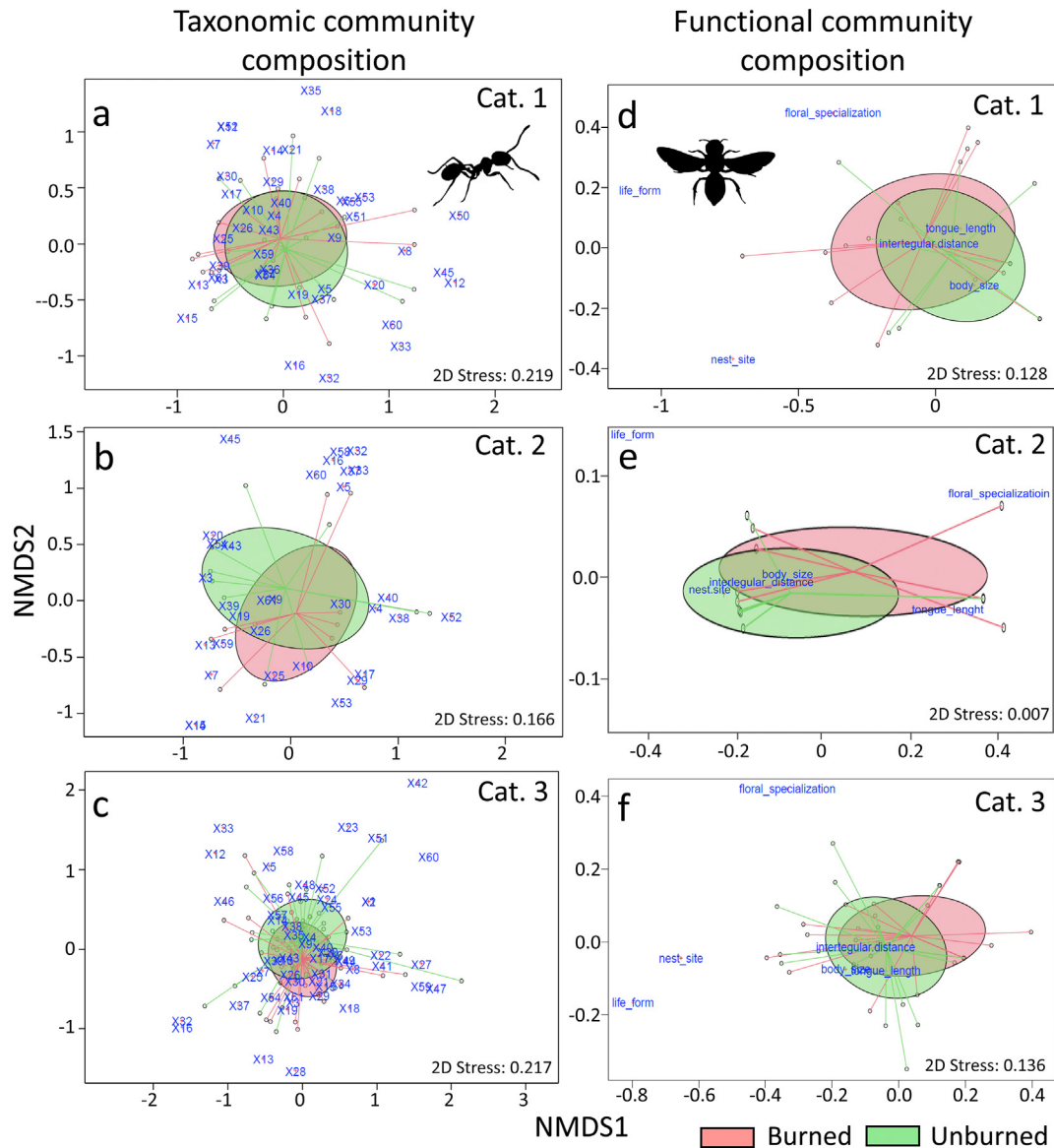


Fig. 6. Non-metric multidimensional scaling analysis of the taxonomic community similarity of ants and the functional community similarity of bees with time since fire. Ant taxonomic similarity at areas that burned (a) recently, (b) moderately recently, and (c) less recently. Bee functional similarity in areas that burned (d) recently, (e) moderately recently, and (f) less recently. In red are the burned areas, and in green are the unburned areas. The 2D stress coefficients are indicated. See species codes in Tables S6, S7.

González-De Vega et al., 2016; Madrigal et al., 2021; Ne'eman et al., 2004). In contrast, in zones that originally hosted forests of *P. nigra* or *P. pinea*, fire created rather open areas with a more or less developed understory. These species do not have serotinous cones. While *P. pinea* can withstand low-intensity fires, thanks to its thick bark, it is unable to resprout after fire-induced mortality (Madrigal et al., 2019). This fact could also explain why, in our study, the ant communities associated with *P. nigra* were significantly more uniform (i.e., had higher values of Pielou's evenness) than the communities associated with *P. pinaster*, *P. halepensis*, and *P. pinea*.

4.4. Do taxonomic and functional composition differ with time since fire?

Our results revealed that the fire, the time since fire, and their interaction had an effect on the taxonomic composition of ant communities. We discovered that ant species with an arboreal lifestyle or an association with forests (e.g., *Camponotus lateralis*, *Crematogaster scutellaris*, *Formica cunicularia*, and various *Temnothorax* species) were more likely to occur at

areas that had burned at least 15 years ago. In these areas, enough time had passed to allow the vegetation to become re-established. Other ant species (e.g., *Messor maroccanus*, *M. lusitanicus*, *Aphaenogaster senilis*, *Cataglyphis floricola*, *C. tartessica*, *C. hispanica*, *Oxyopomyrmex saulcyi*, and *Goniomma kugleri*) were more strongly associated with recent and moderately recent fires, which means the burned area would have less developed vegetation. Such environmental conditions are especially favorable for behaviorally subordinate ant species that live in open areas and that forage at high temperatures when behaviorally dominant ants are absent (Cerdá et al., 1998, 1997). Similar patterns have been seen elsewhere: the bare ground revealed by fire is a boon for thermophilic ant species such as *Forelius pruinosus* in Florida (Izhaki et al., 2003) and *Dorymyrmex* species in Argentina (Sackmann and Farji-Brener, 2006). Our finding that the functional composition was similar could suggest the presence of functional redundancy, where different species can fulfill similar ecosystem functions following a change in environmental conditions, as has been showed in recent works (Arnan et al., 2019; Houadria et al., 2016; Houadria and Menzel, 2017).

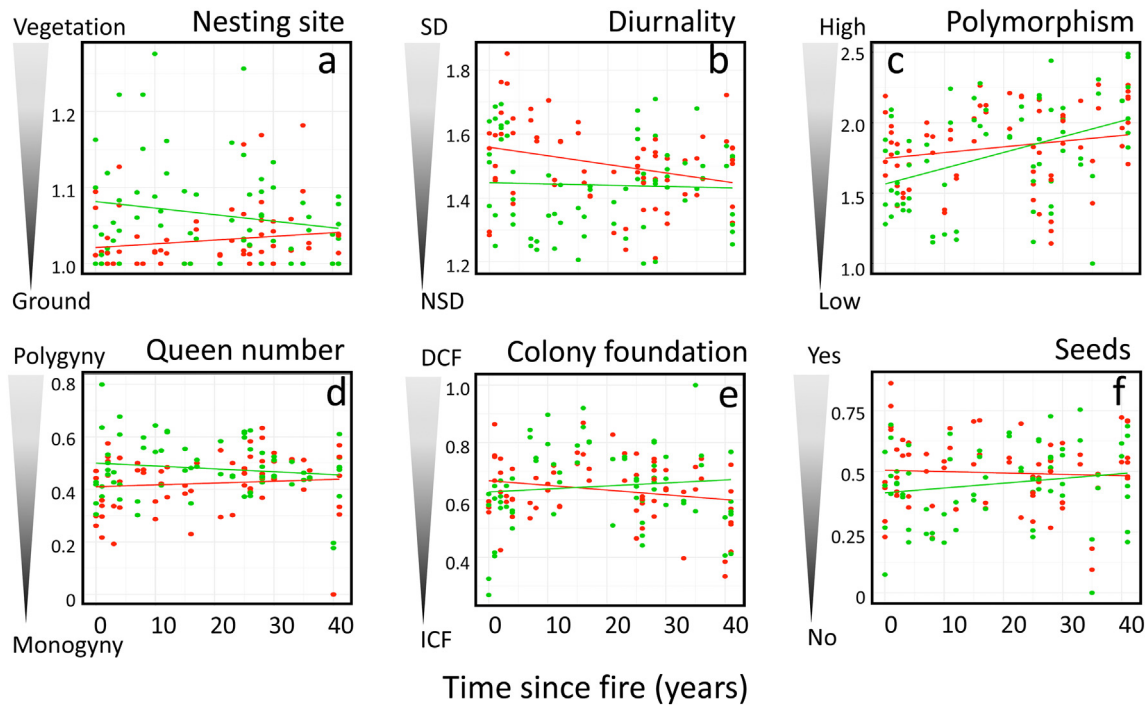


Fig. 7. Effects of fire and time since fire on ant traits: (a) nesting site (ground or vegetation); (b) diurnality (non-strictly diurnal species [NDS] or strictly diurnal species [DS]); (c) degree of worker polymorphism (low, medium, or high); (d) queen number (monogyny, polygyny, or both); (e) seed-based diet (no or yes); (f) colony foundation type (independent colony foundation [ICF], dependent colony foundation [DCF], or both). The burned areas are in red, and the unburned areas are in green. For a detailed description of the traits, see Table S3.

4.5. Which functional traits are filtered by fire?

Several studies have shown that fire broadly shapes terrestrial arthropod communities by causing the local extinction of existing populations and facilitating the arrival of species with a strong dispersal capacity, a tolerance for open areas with little shade, and an ability to exploit the diverse resources available in the post-fire habitat (see a review in Wong et al., 2019). We observed that fire (or the environmental conditions it creates) may act as a trait filter. First, the ant traits most common at recently burned areas promoted survival (ground nesting), persistence (diurnality, monogyny, seed-based diet), and colonization (dependent colony foundation). Second, there were other ant traits related to persistence that were common at the burned areas but that showed no relationship with time since fire (monodomy, individual foraging behaviour, larger body size, and an insect-free diet). These functional traits seem to be advantageous to ants in landscapes resulting from fire disturbance (Arnan et al., 2020; Vidal-Cordero et al., 2022). The fact that some trait differences diminished with time since fire may stem from the fact that vegetation cover develops over time, causing open areas to disappear.

In the case of bees, taxonomic community composition was only affected by the fire, while functional composition of bee communities was affected by the interaction between fire and time since fire. The bees at unburned areas (which had more developed vegetation) were cavity nesters. They create nests in pre-existing holes in trees, tree trunks, and reeds (e.g., *Lithurgus cornutus* and *Chelostoma* species) or vertical galleries in the thick stems of larger plant species (various *Ceratina* species) (Aguado-Martín, 2015; Michener, 2000). At the burned areas, we observed bees that nest on the ground in open and sunny places (e.g., *Amegilla quadrifasciata*, *Halictus smaragdulus*, and *Thyreus* sp., a parasitic species of *Amegilla*) or on tree stumps, such as those that remain after fires (*Xylocopa cantabrita*) (Aguado-Martín, 2015; Michener, 2000). Similar results have been seen for specific functional traits, such as body size, in other forest bee communities (Burkle et al., 2019). Moreover, recently burned areas hosted polylectic bees, which mainly nest on the ground and in woody materials. Such is unsurprising given the greater availability

of herbaceous flowering plants, open sandy areas, and dead wood immediately after fire. Areas that had burned less recently hosted oligolectic species, which makes sense since certain plant species have become dominant and the floral community is not as rich. In addition, the greater amount of vegetation cover increases moisture levels, supporting snail populations and the presence of trees in which hive nesters can establish themselves.

4.6. Limitations and uncertainties

Sampling protocols are sufficiently accurate to provide good quality data about ant and bee community structure and composition after wildfires. However, our study has some limitations. First, we used a paired design to select burned and unburned areas with a space-for-time substitution approach in which unburned areas represented native communities before the fire (Blois et al., 2013; Kelly et al., 2011; Moretti and Legg, 2009). There were large areas (> 4000 ha) of burned *Pinus* forests without fire recurrence (only one fire event) and where the wood was not removed. Moreover, paired control areas had similar slope and orientation characteristics to burned areas. Some components of the fire regime could not be taken into account and could influence the results. For example, fire season, because most of the selected fires were burned along the summer season, when the majority of fires occur; and fire intensity, because this information was not available for all the fires studied. A second issue could be the *Pinus* species. As we have discussed in the Section 4.2, pine species have different fire adaptation strategies (He et al., 2012; Keeley, 2012), and our sites include forests of four different *Pinus* species: *P. pinaster*, *P. halepensis*, *P. pinea* and *P. nigra*, that mainly affect the speed of vegetation recovery. Although, pine species was added as a factor in the analysis performed, it is possible that some bias around the time since fire could remain.

Third, an issue could arise from insect phenology. We sampled during a relatively short time-window (from May through August), and did 48-h of sampling per site, only once in each plot. Although this seemed to be sufficient for ant results (abundances and responses consistent with previous

works), it was not the case for bees (low abundances and responses were hard to understand). Bee abundance was much lower overall than that observed in previous studies (Bogusch et al., 2015; Burkle et al., 2019). Such could reflect the relatively limited understory communities and soil acidification found in conifer forests (Augusto et al., 1998; De Schrijver et al., 2006). Another reason could be that bee communities were only sampled with pan traps, and sometimes too late in the season (end July–August may be too hot for bees, Bosch et al., 1997). This could have biased not only bee richness but also their traits (Wong et al., 2019); although we did not include any trait related to phenology of the bee species. Bee sampling mainly in spring and with a combination of transect walks and pan traps could lead to a more comprehensive characterization of this group (Nielsen et al., 2011; Westphal et al., 2008).

4.7. Implications for management

Assessments of the response of arthropods to disturbances can be used as early indicators of change and inform management and conservation plans for the affected ecosystem (Kremen et al., 1993; Lawes et al., 2005). These may be especially important in global change scenarios where it is crucial to take advantage of synergies between forestry and biodiversity conservation planning, which operate on the same physical space, and even on the same elements.

Our study reveals increased taxonomic richness and diversity and functional divergence for ants regardless of time since fire, suggesting that fire can have a positive effect on ant communities. Therefore, rejuvenation of coniferous systems within forest planning should be considered as a promoter of richness and diversity, not only in ants but also in bees (Gelles et al., 2022). It would be beneficial to develop a long-term effective monitoring system using standardized protocols of arthropods for the detection of ecosystem changes and recovery assessments after wildfires.

5. Conclusions

Our study of how ant and bee communities respond to wildfires over the longer term has allowed us to explore differing patterns in taxonomic diversity, functional diversity, community structure, and community composition. Although we did not clearly identify the factors that influence community recovery, from either a taxonomic or functional perspective, we show that fire acted as a niche-filtering mechanism, with a long-term effect for some traits (such as ant body size) and a short-term for other traits (such as ant- and bee-nesting type or colony foundation). Knowledge about these functional traits can help generate predictions about how species will respond following fire. Future studies should investigate further additional fire characteristics that could impact community recovery, which were not considered in this study, such as fire recurrence, intensity or severity, or other management options of burned areas after fire.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.164132>.

Funding

JMVC was supported by the Caixa-Severo Ochoa International Fellowship Program (2014) (La Caixa Foundation [ID 100010434]; agreement <LCF/BQ/SO14/52250015>). This study received funding from the regional government of Andalusia (2012 Projects of Excellence [RNM 2705] to Jorge Castro, RB, and XC); Programa Operativo FEDER Andalucía 2014–2020, (B-RNM-214-UGR20, DECOPIMED; to Jorge Castro and XC); and the European Regional Development Fund (Interreg V A Spain – Portugal program [POCTEP] through the CILIFO [Ref. 0753_CILIFO_5_E] and FIREPOCTEP [Ref. 0756_FIREPOCTEP_6_E], to XC).

Code availability

The code used for the analyses is available in Appendix Code S1.

Ethics approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

All the authors have seen the manuscript, approved the manuscript, and have consented to its submission for publication.

CRediT authorship contribution statement

J. Manuel Vidal-Cordero: Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing - original draft, Writing - review & editing. **Francisco P. Molina:** Bee identification, Data curation. **Elena Angulo:** Data curation, Formal analysis, Funding acquisition, Writing - review & editing. **Raphaël Boulay:** Conceptualization, Funding acquisition, Methodology, Supervision. **Xim Cerdá:** Conceptualization, Funding acquisition, Writing - review & editing.

Availability of data and material

The data are available in Appendix [Dataset S1](#).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grateful to Dámaris Crespo and Charlotte Hemingway for their help with the field work; William Nairn, M. Carmen Ramírez, Ángela Sabina, Jorge Crespo, and Pablo Gutiérrez for aiding with sample processing; the GIS and Remote Sensing Laboratory (LAST-EBD), and specially David Aragonés, for providing technical support. Anna Turbelin and Jessica Pearce-Duvet provided help editing the ms. Logistic and technical support in Doñana field work was provided by ICTS-RBD-CSIC, Ministry of Science and Innovation and co-financed by FEDER Funds. We thank Fernando Amor for gathering the geological information and José Luis Ruiz (López Neyra Institute of Parasitology and Biomedicine-CSIC) for furnishing statistical advice.

References

- Adams, C.S., Saenz, D., Kidd, K.R., Schalk, C.M., 2022. Disparate patterns of taxonomic and functional predator diversity under different forest management regimes. *Ecol. Indic.* 136, 108591. <https://doi.org/10.1016/j.ecolind.2022.108591>.
- Aguado-Martín, L.O., 2015. *Guía de Campo de los Polinizadores de España* Ediciones (Mundi-Prensa).
- Amor, F., Ortega, P., Jowers, M.J., Cerdá, X., Billen, J., Lenoir, A., Boulay, R.R., 2011. The evolution of worker-queen polymorphism in *Cataglyphis* ants: interplay between individual- and colony-level selections. *Behav. Ecol. Sociobiol.* 65, 1473–1482. <https://doi.org/10.1007/s00265-011-1157-7>.
- Andersen, A.N., Bocciarelli, D., Fairman, R., Radford, I.J., 2014. Conservation status of ants in an iconic region of monsoonal Australia: levels of endemism and responses to fire in the eastern Kimberley. *J. Insect Conserv.* 18, 127–146. <https://doi.org/10.1007/s10841-014-9624-x>.
- Angulo, E., Boulay, R., Ruano, F., Tinaut, A., Cerdá, X., 2016. Anthropogenic impacts in protected areas: assessing the efficiency of conservation efforts using Mediterranean ant communities. *PeerJ* 4, e2773. <https://doi.org/10.7717/peerj.2773>.
- Anjos, D.V., Campos, R.B.F., Ribeiro, S.P., 2015. Temporal turnover of species maintains ant diversity but transforms species assemblage recovering from fire disturbance. *Sociobiology* 62, 389–395. <https://doi.org/10.13102/sociobiology.v62i3.726>.
- Antoine, C.M., Forrest, J.R.K., 2021. Nesting habitat of ground-nesting bees: a review. *Ecol. Entomol.* 46, 143–159. <https://doi.org/10.1111/een.12986>.

- ArcGIS Pro 3.0. . How inverse distance weighted interpolation works <https://pro.arcgis.com/en/pro-app/latest/help/analysis/geostatistical-analyst/how-inverse-distance-weighted-interpolation-works.htm> Accessed 24 January 2023.
- Aman, X., Rodrigo, A., Retana, J., 2006. Post-fire recovery of Mediterranean ground ant communities follows vegetation and dryness gradients. *J. Biogeogr.* 33, 1246–1258. <https://doi.org/10.1111/j.1365-2699.2006.01506.x>.
- Aman, X., Rodrigo, A., Retana, J., 2007. Uncoupling the effects of shade and food resources of vegetation on Mediterranean ants: an experimental approach at the community level. *Ecography* 30, 161–172. <https://doi.org/10.1111/j.0906-7590.2007.04796.x>.
- Aman, X., Rodrigo, A., Molowny-Horas, R., Retana, J., 2009. Ant-mediated expansion of an obligate seeder species during the first years after fire. *Plant Biol.* 12, 842–852. <https://doi.org/10.1111/j.1438-8677.2009.00294.x>.
- Aman, X., Retana, J., Rodrigo, A., Cerdá, X., 2010. Foraging behaviour of harvesting ants determines seed removal and dispersal. *Insect. Soc.* 57, 421–430. <https://doi.org/10.1007/s00040-010-0100-7>.
- Aman, X., Cerdá, X., Retana, J., 2012. Distinctive life traits and distribution along environmental gradients of dominant and subordinate Mediterranean ant species. *Oecologia* 170, 489–500.
- Aman, X., Cerdá, X., Rodrigo, A., Retana, J., 2013. Response of ant functional composition to fire. *Ecography* 36, 1182–1192. <https://doi.org/10.1111/j.1600-0587.2013.00155.x>.
- Aman, X., Cerdá, X., Retana, J., 2017. Relationships among taxonomic, functional, and phylogenetic ant diversity across the biogeographic regions of Europe. *Ecography* 40, 448–457. <https://doi.org/10.1111/ecog.01938>.
- Aman, X., Arcoverde, G.B., Pie, M.R., Ribeiro-Neto, J.D., Leal, I.R., 2018. Increased anthropogenic disturbance and aridity reduce phylogenetic and functional diversity of ant communities in Caatinga dry forest. *Sci. Total Environ.* 631–632, 429–438. <https://doi.org/10.1016/j.scitotenv.2018.03.037>.
- Aman, X., Molowny-Horas, R., Blüthgen, N., 2019. Food resource exploitation and functional resilience in ant communities found in common Mediterranean habitats. *Sci. Total Environ.* 684, 126–135. <https://doi.org/10.1016/j.scitotenv.2019.05.260>.
- Aman, X., Cerdá, X., Rodrigo, A., 2020. Do forest fires make biotic communities homogeneous or heterogeneous? Patterns of taxonomic, functional, and phylogenetic ant diversity at local and regional scales. *Front. For. Glob. Change* 3, 67. <https://doi.org/10.3389/ffgc.2020.00067>.
- Arruda, F.V. de, Teresa, F.B., Layme, V.M.G., Vicente, R.E., Camarota, F., Izzo, T.J., 2022. Fire and flood: how the Pantanal ant communities respond to multiple disturbances? *Perspect. Ecol. Conserv.* 20, 197–204. <https://doi.org/10.1016/j.pecon.2022.04.002>.
- Augusto, L., Bonnaud, P., Ranger, J., 1998. Impact of tree species on forest soil acidification. *For. Ecol. Manag.* 105, 67–78. [https://doi.org/10.1016/S0378-1127\(97\)00270-3](https://doi.org/10.1016/S0378-1127(97)00270-3).
- Azcárate, F.M., Arquer, L., Sánchez, A.M., Peco, B., 2005. Seed and fruit selection by harvester ants, *Messor barbarus*, in Mediterranean grassland and scrubland. *Funct. Ecol.* 19, 273–283. <https://doi.org/10.1111/j.0269-8463.2005.00956.x>.
- Barber, N.A., Lamagdeleine-Dent, K.A., Willand, J.E., Jones, H.P., McCravy, K.W., 2017. Species and functional trait re-assembly of ground beetle communities in restored grasslands. *Biodivers. Conserv.* 26, 3481–3498. <https://doi.org/10.1007/s10531-017-1417-6>.
- Barroso, Á., Amor, F., Cerdá, X., Boulay, R.R., 2013. Dispersal of non-myrmecophilous plants by a “keystone disperser” ant in a Mediterranean habitat reveals asymmetric interdependence. *Insect. Soc.* 60, 75–86. <https://doi.org/10.1007/s00040-012-0268-0>.
- Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. 67. <https://doi.org/10.18637/jss.v067.i01>.
- Bihn, J.H., Gebauer, G., Brandl, R., Bihn, J.H., Gebauer, G., Brandl, R., 2010. Loss of functional diversity of ant assemblages in secondary tropical forests. *Ecology* 91, 782–792.
- Bishop, T.R., Robertson, M.P., van Rensburg, B.J., Parr, C.L., 2014. Elevation-diversity patterns through space and time: ant communities of the Maloti-Drakensberg Mountains of southern Africa. *J. Biogeogr.* 41, 2256–2268. <https://doi.org/10.1111/jbi.12368>.
- Blois, J.L., Williams, J.W., Fitzpatrick, M.C., Jackson, S.T., Ferrier, S., 2013. Space can substitute for time in predicting climate-change effects on biodiversity. *Proc. Natl. Acad. Sci.* 110, 9374–9379. <https://doi.org/10.1073/pnas.1220228110>.
- Bogusch, P., Blažej, L., Trýzna, M., Heneberg, P., 2015. Forgotten role of fires in central European forests: critical importance of early post-fire successional stages for bees and wasps (Hymenoptera: Aculeata). *Eur. J. For. Res.* 134, 153–166. <https://doi.org/10.1007/s10342-014-0840-4>.
- Bosch, J., Retana, J., Cerdá, X., 1997. Flowering phenology, floral traits and pollinator composition in a herbaceous Mediterranean plant community. *Oecologia* 109, 583–591. <https://doi.org/10.1007/s004420050120>.
- Boulay, R., Carro, F., Soriguer, R.C., Cerdá, X., 2007. Synchrony between fruit maturation and effective dispersers' foraging activity increases seed protection against seed predators. *Proc. Royal Soc.* 274, 2515–2522. <https://doi.org/10.1098/rspb.2007.0594>.
- Braga, R.F., Korasaki, V., Andresen, E., Louzada, J., 2013. Dung beetle community and functions along a habitat-disturbance gradient in the Amazon: a rapid assessment of ecological functions associated to biodiversity. *PLoS One* 8, e57786. <https://doi.org/10.1371/journal.pone.0057786>.
- Burkle, L.A., Simanonok, M.P., Durney, J.S., Myers, J.A., Belote, R.T., 2019. Wildfires influence abundance, diversity, and intraspecific and interspecific trait variation of native bees and flowering plants across burned and unburned landscapes. *Front. Ecol. Evol.* 7. <https://doi.org/10.3389/fevo.2019.00252> article 252.
- Cane, J.H., Neff, J.L., 2011. Predicted fates of ground-nesting bees in soil heated by wildfire: thermal tolerances of life stages and a survey of nesting depths. *Biol. Conserv.* 144, 2631–2636. <https://doi.org/10.1016/j.biocon.2011.07.019>.
- Carbone, L.M., Tavela, J., Pausas, J.G., Aguilár, R., 2019. A global synthesis of fire effects on pollinators. *Glob. Ecol. Biogeogr.* 28, 1487–1498. <https://doi.org/10.1111/geb.12939>.
- Castro, F.S., Giovani da Silva, P., Solar, R., Fernandes, G.W., Neves, F.S., 2020. Environmental drivers of taxonomic and functional diversity of ant communities in a tropical mountain. *Insect Conserv. Divers.* 13, 393–403. <https://doi.org/10.1111/icad.12415>.
- Caut, S., Jowers, M.J., Aman, X., Pearce-Duvt, J., Rodrigo, A., Cerdá, X., Boulay, R.R., 2014. The effects of fire on ant trophic assemblage and sex allocation. *Ecol. Evol.* 4, 35–49. <https://doi.org/10.1002/ecs3.714>.
- Cerdá, X., Retana, J., Cros, S., 1997. Thermal disruption of transitive hierarchies in Mediterranean ant communities. *J. Anim. Ecol.* 66, 363–374. <https://doi.org/10.2307/5982>.
- Cerdá, X., Retana, J., Cros, S., 1998. Critical thermal in Mediterranean limits ant species: between mortality risk and foraging performance. *Funct. Ecol.* 12, 45–55. <https://doi.org/10.1046/j.1365-2435.1998.00160.x>.
- Cerdá, X., Aman, X., Retana, J., 2013. Is competition a significant hallmark of ant (Hymenoptera: Formicidae) ecology. *Myrmecol. News* 18 (1), 131–147.
- Costa, F.V., Blüthgen, N., Viana-Junior, A.B., Guerra, T.J., Di Spirito, L., Neves, F.S., 2018. Resilience to fire and climate seasonality drive the temporal dynamics of ant-plant interactions in a fire-prone ecosystem. *Ecol. Indic.* 93, 247–255. <https://doi.org/10.1016/j.ecolind.2018.05.001>.
- Cronin, A.L., Molet, M., Doums, C., Monnin, T., Peeters, C., 2013. Recurrent evolution of dependent colony foundation across eusocial insects. *Annu. Rev. Entomol.* 58, 37–55. <https://doi.org/10.1146/annurev-ento-120811-153643>.
- CSIC-IARA, 1988. Mapa de suelos de Andalucía (E 1:400.000). CSIC-IARA (Junta de Andalucía). Madrid. Descargado en. <https://esdac.jrc.ec.europa.eu/content/mapa-de-suelos-de-andalucia>.
- da Silva, C.V.C., da Silva Goldas, C., Dattilo, W., Dröse, W., de Souza Mendonça Jr., M., Podgaiski, L.R., 2020. Effects of time-since-fire on ant-plant interactions in southern Brazilian grasslands. *Ecol. Indic.* 112, 106094.
- De Schrijver, A., Mertens, J., Geudens, G., Staelens, J., Campforts, E., Luyssaert, S., De Temmerman, L., De Keersmaecker, L., De Neve, S., Verheyen, K., 2006. Acidification of forested podzols in North Belgium during the period 1950–2000. *Sci. Total Environ.* 361, 189–195. <https://doi.org/10.1016/j.scitotenv.2005.06.015>.
- De Vega, C., Herrera, C.M., Dötterl, S., 2014. Floral volatiles play a key role in specialized ant pollination. *Perspect. Plant Ecol. Evol. Syst.* 16, 32–42. <https://doi.org/10.1016/j.ppees.2013.11.002>.
- Del Toro, I., Ribbons, R.R., Pelini, S., 2012. The little things that run the world revisited: a review of ant-mediated ecosystem services and disservices (Hymenoptera: Formicidae). *Myrmecol. News* 17, 133–146 doi:ISSN 1997-3500.
- Elvira, N.J., Lloret, F., Jaime, L., Margalef-Marrase, J., Pérez Navarro, M.Á., Batllori, E., 2021. Species climatic niche explains post-fire regeneration of Aleppo pine (*Pinus halepensis* mill.) under compounded effects of fire and drought in east Spain. *Sci. Total Environ.* 798, 149308. <https://doi.org/10.1016/j.scitotenv.2021.149308>.
- Fagundes, R., Anjos, D.V., Carvalho, R., Del-Claro, K., 2015. Availability of food and nesting-sites as regulatory mechanisms for the recovery of ant diversity after fire disturbance. *Sociobiology* 62, 1–9. <https://doi.org/10.13102/sociobiology.v62i1.1-9>.
- Fagundes, R., Lange, D., Anjos, D.V., de Lima, F.P., Nahas, L., Corro, E.J., ... Dáttilo, W., 2018. Limited effects of fire disturbances on the species diversity and structure of ant-plant interaction networks in Brazilian Cerrado. *Acta Oecol.* 93, 65–73.
- Fontanilla, A.M., Nakamura, A., Xu, Z., Cao, M., Kitching, R.L., Tang, Y., Burwell, C.J., 2019. Taxonomic and functional ant diversity along tropical, subtropical, and subalpine elevational transects in southwest China. *Insects* 10, 128. <https://doi.org/10.3390/insects10050128>.
- Galbraith, S.M., Cane, J.H., Moldenke, A.R., Rivers, J.W., 2019. Wild bee diversity increases with local fire severity in a fire-prone landscape. *Ecosphere* 10, e02668. <https://doi.org/10.1002/ecs2.2668>.
- Gelles, R.V., Davis, T.S., Stevens-Rumann, C.S., 2022. Wildfire and forest thinning shift floral resources and nesting substrates to impact native bee biodiversity in ponderosa pine forests of the Colorado front range. *Forest Ecol. Manag.* 510, 120087. <https://doi.org/10.1016/j.foreco.2022.120087>.
- Gómez, J.M., Zamora, R., Hódar, J.A., García, D., 1996. Experimental study of pollination by ants in Mediterranean High Mountain and arid habitats. *Oecologia* 105, 236–242.
- González-De Vega, S., De las Heras, J., Moya, D., 2016. Resilience of Mediterranean terrestrial ecosystems and fire severity in semiarid areas: responses of Aleppo pine forests in the short, mid and long term. *Sci. Total Environ.* 573, 1171–1177. <https://doi.org/10.1016/j.scitotenv.2016.03.115>.
- Grundel, R., Jean, R.P., Frohnapple, K.J., Glowacki, G.A., Scott, P.E., Pavlovic, N.B., 2010. Floral and nesting resources, habitat structure, and fire influence bee distribution across an open-forest gradient. *Ecol. Appl.* 20, 1678–1692. <https://doi.org/10.1890/08-1792.1>.
- Guédot, C., Bosch, J., Kemp, W.P., 2009. Relationship between body size and homing ability in the genus *Osmia* (Hymenoptera: Megachilidae). *Ecol. Entomol.* 34, 158–161. <https://doi.org/10.1111/j.1365-2311.2008.01054.x>.
- Guiden, P.W., Barber, N.A., Blackburn, R., Farrell, A., Fliginger, J., Hosler, S.C., King, R.B., Nelson, M., Rowland, E.G., Savage, K., Vanek, J.P., Jones, H.P., 2021. Effects of management outreach effects of plant diversity on restored animal communities in tallgrass prairies. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2015421118. <https://doi.org/10.1073/pnas.2015421118>.
- He, T., Pausas, J.G., Belcher, C.M., Schwick, D.W., Lamont, B.B., 2012. Fire-adapted traits of *Pinus* arose in the fiery Cretaceous. *New Phytol.* 194, 751–759. <https://doi.org/10.1111/j.1469-8137.2012.04079.x>.
- Hijmans, R.J., Cameron, S.E., Parra, J.L., Jones, P.G., Jarvis, A., 2005. Very high resolution interpolated climate surfaces for global land areas. *Int. J. Climatol.* 25, 1965–1978. <https://doi.org/10.1002/joc.1276>.
- Houadria, M., Menzel, F., 2017. What determines the importance of a species for ecosystem processes? Insights from tropical ant assemblages. *Oecologia* 184, 885–899. <https://doi.org/10.1007/s00442-017-3900-x>.
- Houadria, M., Blüthgen, N., Salas-Lopez, A., Schmitt, M.I., Amdt, J., Schneider, E., Orivel, J., Menzel, F., 2016. The relation between circadian asynchrony, functional redundancy, and trophic performance in tropical ant communities. *Ecology* 97, 225–235. <https://doi.org/10.1890/14-2466.1>.
- Izhaki, I., Levey, D.J., Silva, W.R., 2003. Effects of prescribed fire on an ant community in Florida pine savanna. *Ecol. Entomol.* 28, 439–448. <https://doi.org/10.1046/j.1365-2311.2003.00528.x>.
- Junta de Andalucía REDIAM, 2022. Áreas recorridas por el fuego en Andalucía (1975-actualidad). <https://portalrediam.cica.es/geonetwork/srv/spa/catalog.search#/metadata/0b9aa872-ad3d-4e30-941a-40a9968b6eb0>.

- Karaman, M.G., 2011. Zoogeography, diversity and altitudinal distribution of ants (Hymenoptera: Formicidae) in the Mediterranean and the oro-Mediterranean parts of Montenegro. *North West J. Zool.* 7, 26–34.
- Keeley, J.E., 2012. Ecology and evolution of pine life histories. *Ann. For. Sci.* 69, 445–453. <https://doi.org/10.1007/s13595-012-0201-8>.
- Keeley, J.E., Pausas, J.G., Rundel, P.W., Bond, W.J., Bradstock, R.A., 2011. Fire as an evolutionary pressure shaping plant traits. *Trends Plant Sci.* 16, 406–411. <https://doi.org/10.1016/j.tplants.2011.04.002>.
- Keeley, J.E., Bond, W.J., Bradstock, R.A., Pausas, J.G., Rundel, P.W., 2012. *Fire in Mediterranean Ecosystems*. Cambridge University Press.
- Kelly, L.T., Nimmo, D.G., Spence-Bailey, L.M., Haslem, A., Watson, S.J., Clarke, M.F., Bennett, A.F., 2011. Influence of fire history on small mammal distributions: insights from a 100-year post-fire chronosequence. *Divers. Distrib.* 17, 462–473. <https://doi.org/10.1111/j.1472-4642.2011.00754.x>.
- Kelly, L.T., Giljohann, K.M., Duane, A., Aquilué, N., Archibald, S., Batllori, E., Bennett, A.F., Buckland, S.T., Canelles, Q., Clarke, M.F., Fortin, M.J., Hermoso, V., Herrando, S., Keane, R.E., Lake, F.K., McCarthy, M.A., Morán-Ordóñez, A., Parr, C.L., Pausas, J.G., Penman, T.D., Regos, A., Rumpff, L., Santos, J.L., Smith, A.L., Syphard, A.D., Tingley, M.W., Brotons, L., 2020. Fire and biodiversity in the Anthropocene. *Science* 370, eabb0355. <https://doi.org/10.1126/science.abb0355>.
- Koh, I., Lonsdorf, E.V., Williams, N.M., Brittain, C., Isaacs, R., Gibbs, J., Ricketts, T.H., 2016. Modeling the status, trends, and impacts of wild bee abundance in the United States. *Proc. Natl. Acad. Sci. U. S. A.* 113, 140–145. <https://doi.org/10.1073/pnas.1517685113>.
- Kremen, C., Colwell, R.K., Erwin, T.L., Murphy, D.D., Noss, R.F., Sanjayan, M.A., 1993. Terrestrial arthropod assemblages: their use in conservation planning. *Conserv. Biol.* 7, 796–808. <https://doi.org/10.1046/j.1523-1739.1993.740796.x>.
- Labrière, E., Legendre, P., 2010. A distance-based framework for measuring functional diversity from multiple traits. *Ecology* 91, 299–305. <https://doi.org/10.1890/08-2244.1>.
- Lawes, M., Kotze, J., Bourquin, S., Morris, C., 2005. Epigeic invertebrates as potential ecological indicators of afro-montane forest condition in South Africa. *Biotropica* 37, 109–118. <https://doi.org/10.1111/j.1744-7429.2005.04054.x>.
- Lazarina, M., Sgardelis, S.P., Tschelin, T., Kallimanis, A.S., Devalze, J., Petanidou, T., 2016. Bee response to fire regimes in Mediterranean pine forests: the role of nesting preference, trophic specialization, and body size. *Basic Appl. Ecol.* 17, 308–320. <https://doi.org/10.1016/j.baae.2016.02.001>.
- López-Baucells, A., Rowley, S., Rocha, R., Bobrowiec, P.E.D., Palmeirim, J.M., Farneda, F.Z., Meyer, C.F.J., 2022. Interplay between local and landscape-scale effects on the taxonomic, functional and phylogenetic diversity of aerial insectivorous neotropical bats. *Landscape Ecol.* 37, 2861–2875. <https://doi.org/10.1007/s10980-022-01493-x>.
- Love, B.G., Cane, J.H., 2016. Limited direct effects of a massive wildfire on its sagebrush steppe bee community. *Ecol. Entomol.* 41, 317–326. <https://doi.org/10.1111/een.12304>.
- Lybrand, R.A., Fedenko, J., Tfaily, M., Rao, S., 2020. Soil properties and biochemical composition of ground-dwelling bee nests in agricultural settings. *Soil Sci. Soc. Am. J.* 84 (4), 1139–1152.
- Madrigal, J., Souto-García, J., Calama, R., Guijarro, M., Picos, J., Hernando, C., 2019. Resistance of *Pinus pinea* L. bark to fire. *Int. J. Wildl. Fire* 28, 342–353. <https://doi.org/10.1071/WF18118>.
- Madrigal, J., Martín, A., Chambel, R., Guijarro, M., Hernando, C., Callejas, M., Espinosa, J., Climent, J., 2021. Do cone age and heating mode determine the opening of serotinous cones during wildfires? A new bench scale approach applied to *Pinus halepensis* mill. *Sci. Total Environ.* 763, 144222. <https://doi.org/10.1016/j.scitotenv.2020.144222>.
- Maechler, M., Rousseeuw, P., Struyf, A., Hubert, M., Hornik, K., 2022. *cluster: Cluster Analysis Basics and Extensions*. R Package Version 2.1.4.
- Májeková, M., Paal, T., Plowman, N.S., Bryndová, M., Kasari, L., Norberg, A., Weiss, M., Bishop, T.R., Luke, S.H., Sam, K., Le Bagousse-Pinguet, Y., Lepš, J., Götzenberger, L., De Bello, F., 2016. Evaluating functional diversity: missing trait data and the importance of species abundance structure and data transformation. *PLoS One* 11, e0149270. <https://doi.org/10.1371/journal.pone.0149270>.
- Mason, N.W.H., Mouillot, D., Lee, W.G., Wilson, J.B., 2005. Functional richness, functional evenness and functional divergence: the primary components of functional diversity. *Oikos* 111, 112–118. <https://doi.org/10.1111/j.0030-1299.2005.13886.x>.
- Mason, S.C., Shirey, V., Ponisio, L.C., Gelhaus, J.K., 2021. Responses from bees, butterflies, and ground beetles to different fire and site characteristics: a global meta-analysis. *Biol. Conserv.* 261, 109265. <https://doi.org/10.1016/j.biocon.2021.109265>.
- McLauchlan, K.K., Higuera, P.E., Miesel, J., Rogers, B.M., Schweitzer, J., Shuman, J.K., Tepley, A.J., Varner, J.M., Veblen, T.T., Adalsteinsson, S.A., Balch, J.K., Baker, P., Batllori, E., Bigio, E., Brando, P., Cattau, M., Chipman, M.L., Coen, J., Crandall, R., Daniels, L., Enright, N., Gross, W.S., Harvey, B.J., Hatten, J.A., Hermann, S., Hewitt, R.E., Kobziar, L.N., Landesmann, J.B., Lortant, M.M., Maezumi, S.Y., Mearns, L., Moritz, M., Myers, J.A., Pausas, J.G., Pellegrini, A.F.A., Platt, W.J., Roozeboom, J., Safford, H., Santos, F., Scheller, R.M., Sherriff, R.L., Smith, K.G., Smith, M.D., Watts, A.C., 2020. Fire as a fundamental ecological process: research advances and frontiers. *J. Ecol.* 108, 2047–2069. <https://doi.org/10.1111/1365-2745.13403>.
- Michener, C.H., 2000. *The Bees of the World*. Johns Hopkins University Press.
- Mola, J.M., Williams, N.M., 2018. Fire-induced change in floral abundance, density, and phenology benefits bumble bee foragers. *Ecosphere* 9, e02056. <https://doi.org/10.1002/ecs2.2056>.
- Mola, J.M., Miller, M.R., O'Rourke, S.M., Williams, N.M., 2020. Wildfire reveals transient changes to individual traits and population responses of a native bumble bee *Bombus vosnesenskii*. *J. Anim. Ecol.* 89, 1799–1810. <https://doi.org/10.1111/1365-2656.13244>.
- Morelli, F., Benedetti, Y., Perna, P., Santolini, R., 2018. Associations among taxonomic diversity, functional diversity and evolutionary distinctiveness vary among environments. *Ecol. Indic.* 88, 8–16. <https://doi.org/10.1016/j.ecolind.2018.01.022>.
- Moretti, M., Legg, C., 2009. Combining plant and animal traits to assess community functional responses to disturbance. *Ecography* 32, 299–309. <https://doi.org/10.1111/j.1600-0587.2008.05524.x>.
- Moretti, M., De Bello, F., Roberts, S.P.M., Potts, S.G., 2009. Taxonomical vs. functional responses of bee communities to fire in two contrasting climatic regions. *J. Anim. Ecol.* 78, 98–108. <https://doi.org/10.1111/j.1365-2656.2008.01462.x>.
- Moretti, M., De Cáceres, M., Pradella, C., Obrist, M.K., Wermelinger, B., Legendre, P., Duelli, P., 2010. Fire-induced taxonomic and functional changes in saproxylic beetle communities in fire sensitive regions. *Ecography* 33, 760–771. <https://doi.org/10.1111/j.1600-0587.2009.06172.x>.
- Mouchet, M.A., Villéger, S., Mason, N.W.H., Mouillot, D., 2010. Functional diversity measures: an overview of their redundancy and their ability to discriminate community assembly rules. *Funct. Ecol.* 24, 867–876. <https://doi.org/10.1111/j.1365-2435.2010.01695.x>.
- Mouillot, D., Graham, N.A.J., Villéger, S., Mason, N.W.H., Bellwood, D.R., 2013. A functional approach reveals community responses to disturbances. *Trends Ecol. Evol.* 28, 167–177. <https://doi.org/10.1016/j.tree.2012.10.004>.
- Ne'eman, G., Dafni, A., Potts, S.G., 2000. The effect of fire on flower visitation rate and fruit set in four core-species in east Mediterranean scrubland. *Plant Ecol.* 146, 97–104. <https://doi.org/10.1023/A:1009815318590>.
- Ne'eman, G., Goubitz, S., Nathan, R., 2004. Reproductive traits of *Pinus halepensis* in the light of fire - a critical review. *Plant Ecol.* 171, 69–79. <https://doi.org/10.1023/B:VEGE.0000029380.04821.99>.
- New, T.R., 2014. *Insects, Fire and Conservation*. Springer, Cham.
- Nielsen, A., Steffan-Dewenter, I., Westphal, C., Messinger, O., Potts, S.G., Roberts, S.P.M., ... Petanidou, T., 2011. Assessing bee species richness in two Mediterranean communities: importance of habitat type and sampling techniques. *Ecol. Res.* 26, 969–983.
- Oksanen, J., Blanchet, F.G., Kindt, R., Legendre, P., Minchin, P.R., O'hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Wagner, H., Oksanen, M.J., 2018. *Vegan: community ecology package*. R Package, Version 2.1.1, pp. 4–6. <https://doi.org/10.1093/molbev/msv334>.
- Ordóñez, J.L., Retana, J., 2004. Early reduction of post-fire recruitment of *Pinus nigra* by post-dispersal seed predation in different time-since-fire habitats. *Ecography* 27, 449–458. <https://doi.org/10.1111/j.0906-7590.2004.03886.x>.
- Parr, C.L., Andersen, A.N., 2008. Fire resilience of ant assemblages in long-unburnt savanna of northern Australia. *Austral Ecol.* 33, 830–838. <https://doi.org/10.1111/j.1442-9993.2008.01848.x>.
- Parr, C.L., Robertson, H.G., Biggs, H.C., Chown, S.L., 2004. Response of African savanna ants to long-term fire regimes. *J. Appl. Ecol.* 41, 630–642. <https://doi.org/10.1111/j.0021-8901.2004.00920.x>.
- Pausas, J.G., 2019. Generalized fire response strategies in plants and animals. *Oikos* 128, 147–153. <https://doi.org/10.1111/oik.05907>.
- Pausas, J.G., Keeley, J.E., 2009. A burning story: the role of fire in the history of life. *Bioscience* 59, 593–601. <https://doi.org/10.1525/bio.2009.59.7.10>.
- Pausas, J.G., Keeley, J.E., 2019. Wildfires as an ecosystem service. *Front. Ecol. Environ.* 17, 289–295. <https://doi.org/10.1002/fee.2044>.
- Pausas, J.G., Parr, C.L., 2018. Towards an understanding of the evolutionary role of fire in animals. *Evol. Ecol.* 32, 113–125. <https://doi.org/10.1007/s10682-018-9927-6>.
- Peralta, G., Stevani, E.L., Chacoff, N.P., Dorado, J., Vázquez, D.P., 2017. Fire influences the structure of plant-bee networks. *J. Anim. Ecol.* 86, 1372–1379. <https://doi.org/10.1111/1365-2656.12731>.
- Perillo, L.N., Barbosa, N.P.D.U., Solar, R.R., Neves, F.D.S., 2020. Patterns of diversity in a metacommunity of bees and wasps of relicual mountainous forest fragments. *J. Insect Conserv.* 24, 17–34. <https://doi.org/10.1007/s10841-019-00194-2>.
- Petanidou, T., Ellis, W.M., 1996. Interdependence of native bee faunas and floras in changing Mediterranean communities. In: Matheson, A., Buchmann, S.L., O'Toole, C., Westrich, P., Williams, I.H. (Eds.), *The Conservation of Bees*, Chapter 16. Academic Press, London, pp. 201–226 1996.
- Pinheiro, J.C., Bates, D.M., 2000. *Mixed-effects models in S and S-PLUS*. Springer, New York. <https://doi.org/10.1007/b98882>.
- Pinheiro, J.C., Bates, D.M., R. Core Team, 2022. *nlme: Linear and Nonlinear Mixed Effects Models*. R package version 3, pp. 1–160. <https://CRAN.R-project.org/package=nlme>.
- Potts, S.G., Vulliamy, B., Dafni, A., Ne'eman, G., O'Toole, C., Roberts, S., Willmer, P., 2003. Response of plant-pollinator communities to fire: changes in diversity, abundance and floral reward structure. *Oikos* 101, 103–112.
- Prendergast, K.S., Menz, M.H.M., Dixon, K.W., Bateman, P.W., 2020. The relative performance of sampling methods for native bees: an empirical test and review of the literature. *Ecosphere* 11, e03076.
- R Core Team, 2022. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria URL <https://www.R-project.org/>.
- Rader, R., Bartomeus, I., Tylianakis, J.M., Labrière, E., 2014. The winners and losers of land use intensification: pollinator community disassembly is non-random and alters functional diversity. *Divers. Distrib.* 20, 908–917. <https://doi.org/10.1111/ddi.12221>.
- Retana, J., Cerdá, X., 2000. Patterns of diversity and composition of Mediterranean ground ant communities tracking spatial and temporal variability in the thermal environment. *Oecologia* 123, 436–444. <https://doi.org/10.1007/s004420051031>.
- Retana, J., Pico, F.X., Rodrigo, A., 2004. Dual role of harvesting ants as seed predators and dispersers of a non-myrmecophilous Mediterranean perennial herb. *Oikos* 105, 377–385.
- Retana, J., Aman, X., Cerdá, X., 2015. A multidimensional functional trait analysis of resource exploitation in European ants. *Ecology* 96, 2781–2793. <https://doi.org/10.1890/14-2326.1>.
- Ricotta, C., Moretti, M., 2011. CWM and Rao's quadratic diversity: a unified framework for functional ecology. *Oecologia* 167, 181–188. <https://doi.org/10.1007/s00442-011-1965-5>.
- Ridel, A., Lafage, D., Devogel, P., Lacoue-Labarthe, T., Pétillon, J., 2021. Habitat filtering differentially modulates phylogenetic and functional diversity relationships between predatory arthropods. *R. Soc. Open Sci.* 8, 202093. <https://doi.org/10.1098/rsos.202093>.
- Rodrigo, A., Retana, J., 2006. Post-fire recovery of ant communities in submediterranean *Pinus nigra* forests. *Ecography* 29, 231–239. <https://doi.org/10.1111/j.2006.0906-7590.04272.x>.

- Rosa, T.F., Camarota, F., Zuanon, L.A., Tito, R., Maravalhas, J.B., Powell, S., Vasconcelos, H.L., 2021. The effects of high-severity fires on the arboreal ant community of a Neotropical savanna. *Oecologia* 196 (4), 951–961.
- Sackmann, P., Farji-Brener, A., 2006. Effect of fire on ground beetles and ant assemblages along an environmental gradient in NW Patagonia: does habitat type matter? *Ecoscience* 13, 360–371. <https://doi.org/10.2980/i1195-6860-13-3-360.1>.
- Sanders, N.J., 2002. Elevational gradients in ant species richness: area, geometry, and Rapoport's rule. *Ecography* 25, 25–32. <https://doi.org/10.1034/j.1600-0587.2002.250104.x>.
- Shik, J.Z., Arnan, X., Oms, C.S., Cerdá, X., Boulay, R., 2019. Evidence for locally adaptive metabolic rates among ant populations along an elevational gradient. *J. Anim. Ecol.* 88, 1240–1249. <https://doi.org/10.1111/1365-2656.13007>.
- Simmons, S.A., Bossart, J.L., 2020. Apparent resilience to fire of native bee (Hymenoptera: Apoidea) communities from upland longleaf pine forests in Louisiana and Mississippi. *Southeast. Nat.* 19, 567–581. <https://doi.org/10.1656/058.019.0316>.
- Suggitt, A.J., Stefanescu, C., Páramo, F., Oliver, T., Anderson, B.J., Hill, J.K., Roy, D.B., Brereton, T., Thomas, C.D., 2012. Habitat associations of species show consistent but weak responses to climate. *Biol. Lett.* 8, 590–593. <https://doi.org/10.1098/rsbl.2012.0112>.
- Verble-Pearson, R.M., Yanoviak, S.P., 2014. Effects of fire intensity on litter arthropod communities in Ozark oak forests, Arkansas, U.S.A. *Am. Midl. Nat.* 172, 14–24. <https://doi.org/10.1674/0003-0031-172.1.14>.
- Vidal-Cordero, J.M., Aman, X., Rodrigo, A., Cerdá, X., Boulay, R., 2022. Four-year study of arthropod taxonomic and functional responses to a forest wildfire: Epigeic ants and spiders are affected differently. *For. Ecol. Manag.* 520, 120379. <https://doi.org/10.1016/j.foreco.2022.120379>.
- Viljur, M.L., Abella, S.R., Adámek, M., Alencar, J.B.R., Barber, N.A., Beudert, B., Burkle, L.A., Cagnolo, L., Campos, B.R., Chao, A., Chergui, B., Choi, C.Y., Cleary, D.F.R., Davis, T.S., Dechnik-Vázquez, Y.A., Downing, W.M., Fuentes-Ramirez, A., Gandhi, K.J.K., Gehring, C., Georgiev, K.B., Gimbutas, M., Gongalsky, K.B., Gorbunova, A.Y., Greenberg, C.H., Hylander, K., Jules, E.S., Korobushkin, D.I., Köster, K., Kurth, V., Lanham, J.D., Lazarina, M., Leverkus, A.B., Lindenmayer, D., Marra, D.M., Martín-Pinto, P., Meave, J.A., Moretti, M., Nam, H.Y., Obrist, M.K., Petanidou, T., Pons, P., Potts, S.G., Rapoport, I.B., Rhoades, P.R., Richter, C., Saifutdinov, R.A., Sanders, N.J., Santos, X., Steel, Z., Tavela, J., Wendenburg, C., Wermelinger, B., Zaitsev, A.S., Thorn, S., 2022. The effect of natural disturbances on forest biodiversity: an ecological synthesis. *Biol. Rev.* 97, 1930–1947. <https://doi.org/10.1111/brv.12876>.
- Villalta, I., Bouget, C., Lopez-Vaamonde, C., Baude, M., 2022. Phylogenetic, functional and taxonomic responses of wild bee communities along urbanisation gradients. *Sci. Total Environ.* 832, 154926. <https://doi.org/10.1016/j.scitotenv.2022.154926>.
- Villéger, S., Ramos Miranda, J., Flores Hernández, D., Mouillot, D., 2010. Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecol. Appl.* 20, 1512–1522. <https://doi.org/10.1890/09-1310.1>.
- Ward, D.F., New, T.R., Yen, A.L., 2001. Effects of pitfall trap spacing on the abundance, richness and composition of invertebrate catches. *J. Insect Conserv.* 5, 47–53. <https://doi.org/10.1023/A:1011317423622>.
- Wermelinger, B., Moretti, M., Duelli, P., Lachat, T., Pezzatti, G.B., Obrist, M.K., 2017. Impact of windthrow and salvage-logging on taxonomic and functional diversity of forest arthropods. *For. Ecol. Manag.* 391, 9–18. <https://doi.org/10.1016/j.foreco.2017.01.033>.
- Westphal, C., Bommarco, R., Carré, G., Lamborn, E., Morison, N., Petanidou, T., Potts, S.G., Roberts, S.P.M., Szentgyörgyi, H., Tscheulin, T., Vaissière, B.E., Woyciechowski, M., Biesmeijer, J.C., Kunin, W.E., Settele, J., Steffan-Dewenter, I., 2008. Measuring bee diversity in different European habitats and biogeographical regions. *Ecol. Monogr.* 78, 653–671. <https://doi.org/10.1890/07-1292.1>.
- Wong, M.K.L., Guénard, B., Lewis, O.T., 2019. Trait-based ecology of terrestrial arthropods. *Biol. Rev.* 94, 999–1022. <https://doi.org/10.1111/brv.12488>.
- Yekwayo, I., Pryke, J.S., Gaigher, R., Samways, M.J., 2018. Only multi-taxon studies show the full range of arthropod responses to fire. *PLoS One* 13 (4), e0195414. <https://doi.org/10.1371/journal.pone.0195414>.